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Medical AI Agents: A Comprehensive Survey of Architectures, Cognitive Modules, and Clinical Workflows

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Abstract—Agent-driven multi-agent systems, built upon large language models (LLMs), are emerging as a promising paradigm to address the challenges of complex data and accelerated decision-making in medicine. However, despite rapid progress, research in this area remains fragmented across narrow scenarios, making it difficult to systematically assess and adopt these technologies. To address this gap and synthesize the fragmented landscape, this survey introduced a task-oriented taxonomy that organizes medical AI agents into four core application landscapes: Clinical natural language processing, Medical Computer Vision, Graph Learning, and Omics and Biology. Drawing on recent studies published mainly between 2023 and 2025, this review characterized the roles that agents serve for clinicians and patients and discussed how existing systems integrate reasoning, memory, and tool use into coherent pipelines aligned with concrete clinical workflows. The availability of code, models, and agent implementations is also documented to inform practical reuse. Furthermore, the review synthesized cross-cutting challenges reported in the literature, including data quality, clinical grounding, safety, transparency, and deployment. Finally, it outlined future directions for building robust, controllable, and clinically actionable medical AI agents.

Index Terms—AI Agents, Large Language Models, Medical AI, Multi-agent systems

I. INTRODUCTION

AI agents are emerging as a transformative paradigm in medicine, a field confronted by two critical, interconnected challenges: the proliferation of heterogeneous, multi-modal data and the pressing need to accelerate biomedical discovery and clinical decision-making. Building on advances in large language models such as GPT, PaLM, Llama, and Qwen [1]–[5], a new generation of AI agents has emerged as a promising paradigm to address these demands [6]–[11]. By leveraging LLMs as a core cognitive engine [12] and integrating modules for planning [13], memory [14], retrieval, and tool use [15], these systems extend far beyond static prediction. They enable agents capable of decomposing complex clinical tasks, orchestrating specialized tools and knowledge sources, and adapting their reasoning to dynamically evolving medical scenarios.

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AI agents are being actively explored across key facets of medicine. In clinical care, they support information extraction from electronic health records (EHRs), clinical documentation, multi-turn triage dialogues, and decision support [16], [17]. In medical imaging, especially radiology, agentic systems couple visual encoders with reasoning and reporting modules to localize findings and draft radiology reports from radiographs and cross-sectional imaging [18], [19]. In the realm of biomedical research, agents orchestrate complex processes including scientific literature mining, knowledge graphs construction, and omics analysis to accelerate target discovery, drug design, and biomarker identification [20], [21]. Together, these developments mark a shift from standalone predictive models toward agentic co-pilots that collaborate with clinicians and scientists in complex workflows [22].

Despite rapid progress, research on medical AI agents remains fragmented across highly specialized scenarios, complicating both principled system design for developers and hindering practical adoption by medical professionals. Existing surveys largely mirror this fragmentation: some concentrate on internal architectures and mechanisms [23], [24], others focus on evaluation environments and simulated workflows [25]–[27], and others primarily curate datasets, tools, or benchmarks [28]. Although these perspectives are valuable but rarely provide a unified view that (i) links concrete clinical problems to specific agent components, (ii) spans core domains such as clinical natural language processing (NLP), medical imaging, graph learning, and omics, and (iii) connects these elements to implementations that can be readily reused in practice.

Recent publication trends further highlight this fragmentation alongside the field’s momentum. As shown in Fig. 1, we analyze both arXiv and PubMed records from 2020 to 2025 to characterize the evolution of medical AI agent research. The arXiv curve exhibits a sharp surge beginning in 2023, reflecting increasing methodological innovation and the rapid emergence of agentic frameworks built upon large language models. In contrast, the trend in PubMed reveals a domain-structured growth pattern: clinical NLP agents expand most rapidly, while medical vision agents, graph-based agents, and omics-driven agents show steady but uneven development. Together, these complementary trends reveal both an accelerating research trajectory and a lack of unified cross-domain organization, motivating the need for an integrated, task-oriented survey.

As Table I illustrates, a comparison of representative surveys confirms this fragmentation, showing that existing works do not simultaneously cover the core pillars of agent architecture, multi-domain disciplinary depth, and clinical workflow mapping, and resource availability.

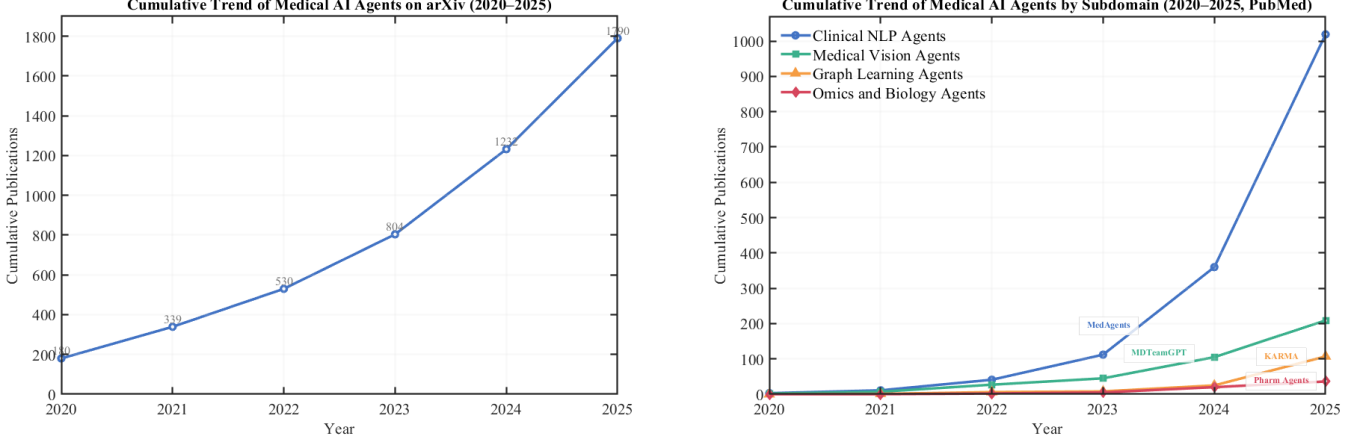


Fig. 1. Cumulative trends of medical AI agents from 2020 to 2025 on arXiv and PubMed. The arXiv trend shows a rapid surge in medical-focused AI agent publications since 2023, with growth accelerating further in 2024–2025. The PubMed trend breaks down cumulative publications by subdomain, revealing that clinical NLP agents dominate the increase, while vision, graph, and omics agents exhibit more gradual growth.

In this survey, we address this gap with a task-oriented taxonomy grounded in real clinical and research needs. Our primary contributions are as follows:

- We introduce a novel, task-oriented taxonomy that systematically organizes the entire domain of medical AI agents into four pivotal application landscapes: clinical natural language processing, advanced medical computer vision, sophisticated graph learning, and intricate omics and biology.
- For each domain, we characterize the roles that agents serve for clinicians and patients, describe how existing systems integrate reasoning, memory, retrieval, and tool use into coherent pipelines, and align these designs with concrete clinical workflows.
- We document whether each study releases code, models, or agent implementations to provide practical entry points for adaptation and deployment in real-world settings.
- By comparing patterns across domains, we distill cross-cutting challenges in data quality, clinical grounding, safety, transparency, and deployment, and outline directions for building robust, controllable, and clinically actionable medical AI agents.
- We quantify publication trends in medical AI agents on arXiv and PubMed from 2020 to 2025 and perform an in-depth synthesis of representative systems mainly published between 2023 and 2025, providing a timely and comprehensive snapshot of this rapidly evolving field.

The remainder of this paper is organized as follows. Fig. 3 providing a detailed structural roadmap for the entire survey. Section II reviews the foundational background of AI agents, detailing their core cognitive modules, including foundational models (Section II-A), planners (Section II-B), memory (Section II-C), and tools (Section II-D). Section III introduces and elaborates on our novel, task-oriented taxonomy, which forms the core of our survey. This section systematically analyzes the four pivotal application landscapes: Clinical NLP (Section III-A), Medical Computer Vision (Section III-B), Graph

Learning (Section III-C), and Omics and Biology (Section III-D). Building on this taxonomic review, Section IV distills the cross-cutting challenges that hinder current adoption, and Section V outlines future research directions for building robust and controllable medical AI agents. Section VI provides a conclusion that summarizes the primary contributions of this survey.

II. BACKGROUND

A medical AI agent is not a single model but a complete system composed of multiple synergistic components. As illustrated in Fig. 2, a typical medical AI agent operates as an end-to-end pipeline rather than an isolated predictor. The cycle begins with perception, where multimodal clinical data (text, images, omics, and structured EHR) are encoded into shared representations, and memory, which maintains short- and long-term patient and knowledge states. Based on these representations, the decision module performs planning and reasoning, orchestrating retrieval, tool calls, and knowledge synthesis to generate candidate actions and recommendations. Finally, the Evaluation stage incorporates reflection, human-in-the-loop verification, and knowledge updates to close the loop and continuously improve the agent’s behavior.

In the remainder of this section, we decompose this pipeline into its core building blocks, namely foundational models (Section II-A), planners (Section II-B), memory (Section II-C), and tools (Section II-D). We also summarize how these components are instantiated in current medical AI agents.

A. Foundational Models

The foundation model lies at the core of an AI agent [29], which serves as its computational “brain” for reasoning and decision-making. These are typically LLMs such as GPT-4, PaLM or multimodal large models, whose development has significantly accelerated the progress of AI agents. These models provide essential capabilities for understanding language, interpreting multi-modal information, and interacting

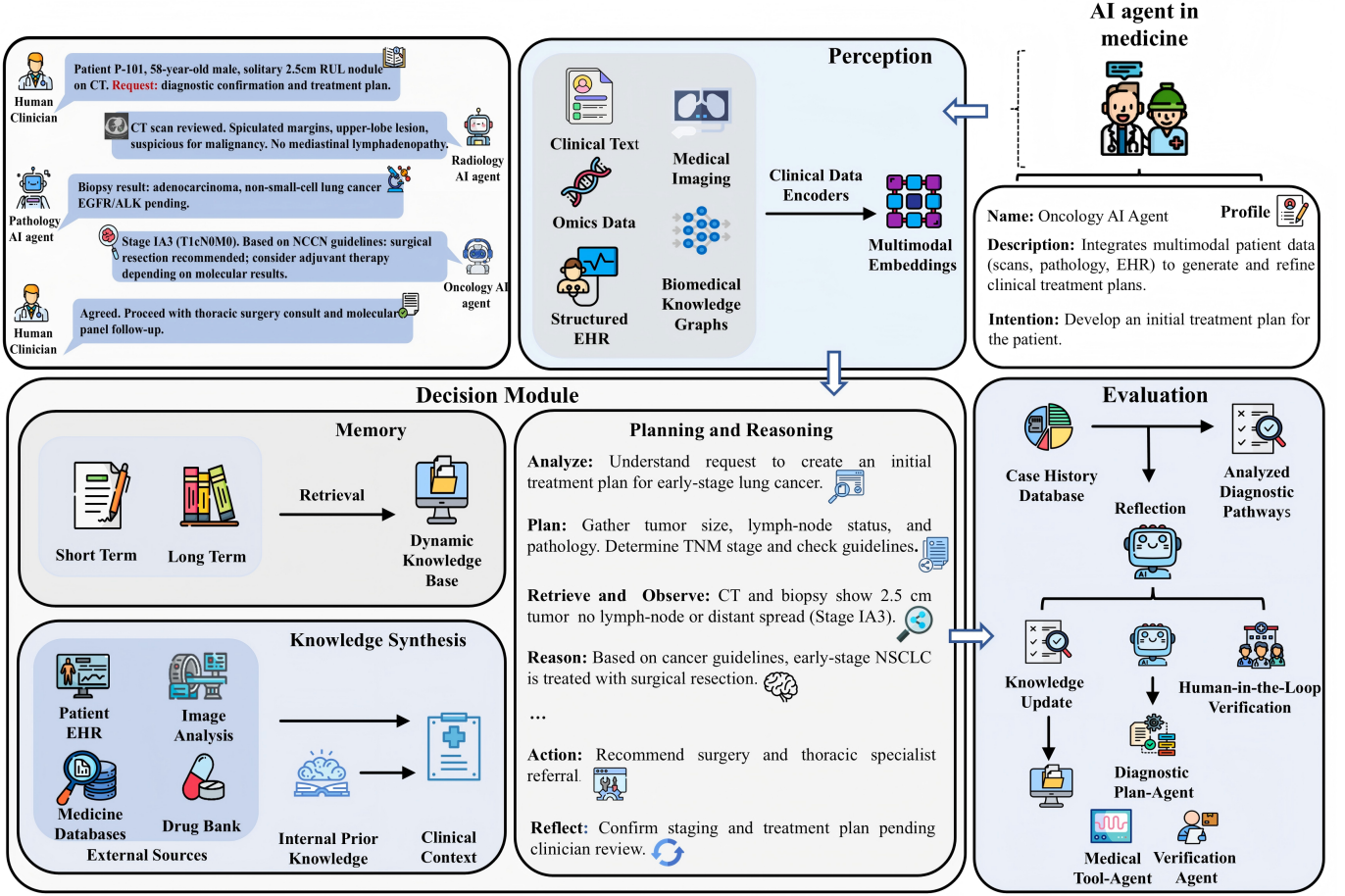


Fig. 2. Illustrative example of a medical AI agent workflow in a multidisciplinary lung cancer case. The figure highlights typical components, including multimodal perception, memory and knowledge synthesis, planning and reasoning, and evaluation with human in the loop verification, to show how agentic modules can be organized to support complex clinical decision processes.

intelligently with their environments [30]. Specifically, in the medical field, specialized LLMs like MedAlpaca [31] and Med-Palm [32] further enhance diagnostic reasoning, clinical communication, and workflow automation. As the computational “brain” of medical agents, these models entitle specialized agents to enhance diagnostic accuracy, improve the efficiency of clinical workflows, and refine patient interactions.

B. Planner

The planner module acts as the logical core of AI agents, responsible for translating high-level goals into sub-tasks [30]. Instead of merely composing tasks in sequence, the Planner autonomously decomposes complex challenges into sub-tasks and coordinates with other modules. This capability is vital for medical applications, as an effective medical planning system can significantly enhance the overall performance of clinical AI agents and reduce the likelihood of errors [33].

C. Memory

Memory enables AI agents to store, retrieve, and apply medical information over time, extending far beyond simple context recording [34]. A memory module allows an agent to maintain a detailed historical context of patient interactions

and integrate external knowledge from sources like medical knowledge graphs and databases. This ensures the accuracy and consistency of clinical decisions, closely mirroring the cognitive processes of a human specialist. Furthermore, a robust memory system is crucial for maintaining coherence in extended medical dialogues, retrieving relevant knowledge on demand, and adapting behavior based on past experiences [35]. Given its growing importance for complex medical reasoning, this survey summarizes the application of memory modules across various medical agents.

D. Tools

The Tools module equips AI agents with specialized functionalities, analogous to how humans use instruments to solve complex tasks [36]. This capability allows agents to extend their native abilities, simplifying problem-solving and improving efficiency. In medical practice, agents leverage tools such as image analysis software for interpreting scans, EHR interfaces for patient data management, and drug interaction checkers to ensure medication safety [37]. They can also query evidence-based medicine databases to ground diagnostic and treatment decisions in established knowledge. By integrating these specialized tools, agents significantly enhance their re-

TABLE I
COMPARISON OF REPRESENTATIVE SURVEYS ON MEDICAL AI AGENTS IN TERMS OF ARCHITECTURE, DISCIPLINARY DEPTH, WORKFLOW ALIGNMENT, AND AVAILABLE RESOURCES.

Survey	Architecture & Components	Disciplinary Depth	Clinical Workflow Mapping	Resources / Datasets
Mehandru et al. (2024) [25]	✗	✗	✓	✗
Fan et al. (2024) [26]	✓	✗	✓	✓
Wang et al. (2025) [35]	✓	✗	✗	✗
Yao & Yu (2025) [23]	✓	✗	✓	✗
Chen et al. (2025) [28]	✗	✗	✗	✓
Jiang et al. (2025) [27]	✗	✗	✓	✓
Our Survey	✓	✓	✓	✓

liability, accuracy, and practical utility in real-world clinical settings [22].

III. PROPOSED TAXONOMY

This section elaborates on our proposed taxonomy for medical AI agents. Fig. 3 serves as the detailed structural roadmap for this chapter and the entire survey, outlining the four core domains: Clinical NLP, Medical Computer Vision, Graph Learning, and Omics and Biology. It further breaks down the sub-topics explored within each domain, such as the specific Domain, Agent roles, Tooling and Memory, and Use cases. To support this analysis, we present a series of comprehensive tables (Tables II-VI) that summarize key studies published between 2023 and 2025. These tables deconstruct each agent framework, detailing its core task, pipeline design, knowledge handling strategy, and supported data modalities, thereby providing a structured and detailed inventory of the current research landscape. Fig. 4 provides a high-level visual summary of the proposed taxonomy, centering on the four key domains and detailing each quadrant’s Core Objective, Key Applications, Key Modalities, and representative agent models analyzed in this survey.

A. Clinical NLP

This section delves into AI agents within the Clinical NLP domain. Table II summarizes representative frameworks in this field, detailing their tasks, pipeline designs, knowledge handling strategies, and modalities. Some frameworks (e.g., AIPatient, KG4Diagnosis, MedAssist) combine clinical text with structured KGs; here we classified them under Clinical NLP when the main interface is textual dialogue or reports, and revisit their graph aspects in Sec. III-C.

1) *Domain*: Research in Clinical NLP has primarily evolved along two complementary directions: Clinical Dialogue Systems and Clinical Text Analysis. The former emphasizes interactive, multi-agent communication to support decision-making, while the latter focuses on extracting and structuring knowledge from unstructured clinical text. Together, they form the foundation for intelligent, collaborative, and interpretable clinical applications. In this survey, we categorize a framework under Clinical NLP whenever clinical text, dialogue, or report generation serves as the primary interface for reasoning and interaction, even if the system additionally incorporates structured data, knowledge graphs, or other modalities as supporting signals.

a) *Clinical Dialogue Systems*: Clinical Dialogue Systems leverage the capabilities of LLMs to create intelligent agents capable of engaging in collaborative multi-round discussions. This field has evolved from early single-agent chatbots to a current paradigm focused on multi-agent collaboration. Pioneering this approach, Tang et al. [16] introduced a framework that uses expert role-playing to enable explicit reasoning and provide high-quality dialogue support for clinical inquiries. Building on this, recent work has introduced more dynamic and adaptive designs; for instance, Kim et al. [38] proposed a system that dynamically assigns an optimal collaboration strategy. This paradigm has also been applied to structured data, with ColaCare [39] integrating multi-agent collaboration to enhance the interpretability and performance of predictions from EHR data. Overall, the multi-agent collaboration paradigm represents a significant evolution in the field, leveraging a human-like approach to effectively enhance knowledge utilization and faithfulness in clinical dialogue systems.

b) *Clinical Text Analysis*: Clinical Text Analysis focuses on tasks like information extraction and knowledge graph construction, serving as a crucial step for transforming unstructured clinical text into a structured, machine-readable format. Initial approaches focused on creating static knowledge bases from text. For instance, Wang et al. [39] exemplify this progression with the ColaCare system, which combines expert models for processing structured EHR data with LLMs to generate human-understandable reports and synthesize clinical evidence. These principles are also being applied to automate critical clinical workflows. For example, the TRIAGEAGENT [40] is designed for clinical triage, performing zero-shot document classification. These advancements in Clinical Text Analysis demonstrate the potential of AI-based agent frameworks to transform unstructured medical data into structured knowledge.

2) *Agent roles*: In clinical NLP, multi-agent systems are designed to mirror and augment real-world healthcare dynamics. The agents within these systems can be fundamentally categorized by their primary beneficiary and function: those that emulate the patient and those that assist the clinician. This dichotomy—patient-centered agents and clinician-centered agents—forms the cornerstone for constructing realistic, interactive, and trustworthy clinical simulation and decision-support environments.

a) *Patient-centered Agents*: Patient-centered agents are sophisticated constructs designed to simulate wide spectrum

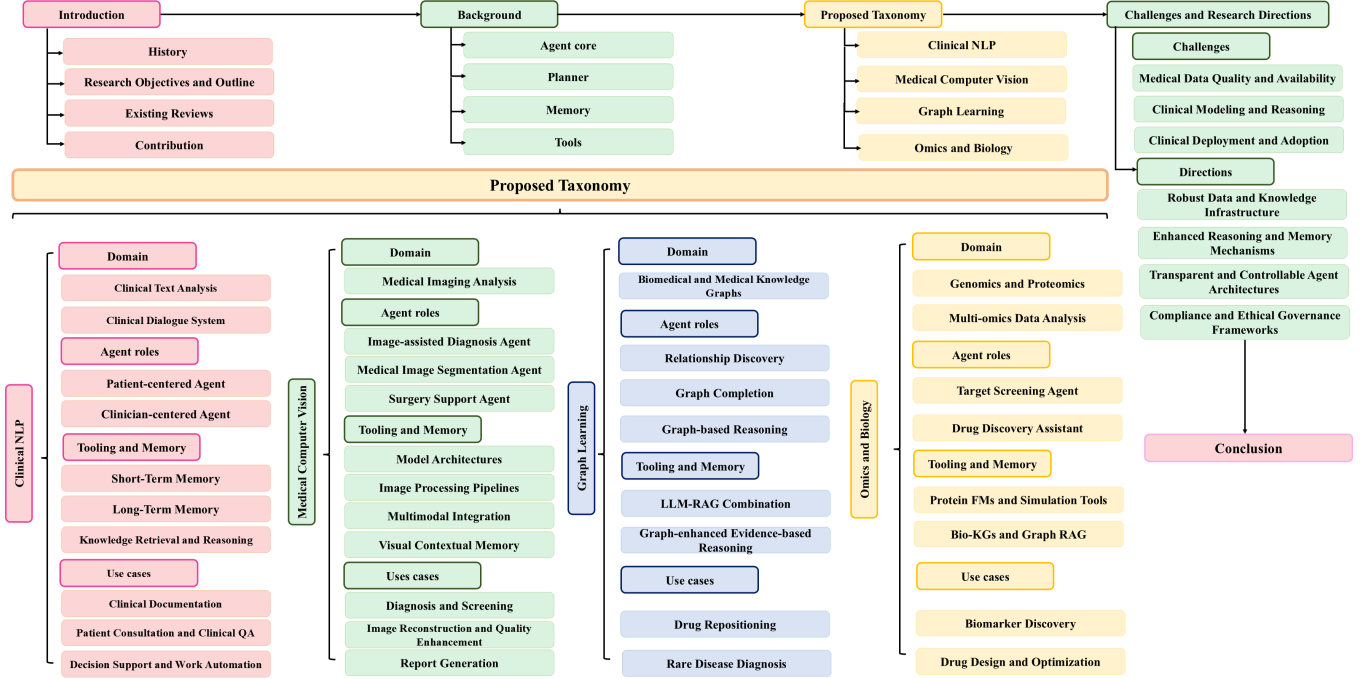


Fig. 3. An overview of the survey’s structure, centered on the proposed task-oriented taxonomy. The framework flows from the Background of core agent components into the four main taxonomic domains: Clinical NLP, Medical Computer Vision, Graph Learning, and Omics and Biology. The lower half details the fine-grained breakdown for each domain, including its specific Domain, Agent roles, Tooling and Memory, and Use cases. The structure concludes by outlining the field’s key Challenges and Open research Directions.

of patient phenotypes, histories, and interactive behaviors. Their evolution has moved far beyond static, scripted responses to dynamic entities that leverage techniques like prompt engineering, Retrieval-Augmented Generation (RAG), and knowledge graphs to generate realistic and contextually coherent patient narratives. For instance, frameworks such as AIPatient [41] leverage real-world data like EHRs and dedicated knowledge graphs to build patient agents that provide realistic histories and maintain conversational continuity. This data-driven approach is further extended by simulators that learn from real dialogue strategies, as seen in the work by Liu et al. [42], this design enables human-like behavior and improves realism.

The applications of patient-centered agents are diverse and impactful. In medical education, frameworks like EvoPatient [43] utilize patient agents in multi-agent systems to simulate collaborative clinical environments, allowing medical students to practice crucial skills like inquiry and teamwork. In tackling complex diagnostic challenges, such as rare diseases, frameworks like RareAgents [44] utilize a patient agent to interact with a multi-disciplinary team of specialist agents, testing the system’s ability to handle nuanced and complex cases. The refinement of these interactions is a key area of innovation, with systems like DoctorAgent-RL [45] using patient agents in a reinforcement learning loop to optimize the doctor’s questioning strategies, while MDTeamGPT [46] uses self-evolving mechanisms to improve diagnostic rationality. Together, these advancements highlight the pivotal role of patient-centered agents in pushing the boundaries of medical AI, from enhancing diagnostic accuracy to improving clinical

training and collaborative care.

b) Clinician-centered Agents: Clinician-centered agents represent a paradigm shift in medical AI, transitioning from passive information retrievers to active collaborators that simulate expert reasoning, facilitate teamwork, and orchestrate complex clinical workflows, mirroring the multi-specialist nature of modern healthcare. Systems like AMSC [47] models hierarchical teams of general practitioners and specialists for in-depth diagnosis. Other frameworks, including MDAgents [38] and ClinicalLab [48], dynamically adjust team size based on task complexity, mirroring real-world clinical practice. To reach reliable consensus, frameworks like MEDAGENTS [16], ColaCare [39], and MoodAngels [49] enable agents to engage in multi-round debates, while Ke et al. [50] explicitly explore this mechanism for reducing cognitive bias.

Beyond collaboration, a significant role for these agents is providing robust clinical decision support. Some frameworks like MD2GPS [51], simulate expert debate to mitigate diagnostic biases, while others, such as openCHA [52], coordinate external tools and data to deliver personalized, evidence-based responses. Rigorous evaluation is also embedded within these systems: Discuss-RAG [53] deploy critic agents to scrutinize preliminary diagnoses, while DoctorAgent-RL [17] refine reasoning through reinforcement and evaluator feedback. Modular designs such as MEDDxAgent [54] unify these strategies, leveraging iterative reasoning to enhance reliability.

Furthermore, clinician-centered agents extend to modeling doctor-patient interaction from the clinician’s perspective, creating rich diagnostic environments for training and benchmarking. AI Hospital [55] benchmarks full diagnostic workflows

with LLM-driven doctors, while NoteChat [56] and Chen et al. [57] generate realistic consultation dialogues for medical education. A final critical role is the automation of specialized procedural workflows, acting as professional assistants in narrow yet complex domains. For example, TRIAGEAGENT [40] focuses on automating guideline-grounded triage level prediction to support referral and prioritization decisions.

In essence, clinician-centered agents extend beyond simple question answering by offering collaborative decision support, simulating clinician–patient interactions, orchestrating complex workflows, and embedding rigorous validation mechanisms, thereby directly enhancing diagnostic accuracy and clinical efficiency. They are engineered to directly enhance diagnostic accuracy, improve clinical efficiency, and integrate as trustworthy partners within the healthcare team.

3) *Tooling and Memory*: Multi-agent systems rely on memory and tool usage to extend reasoning capabilities and improve clinical adaptability. Memory is typically categorized into short-term memory, which supports dynamic interaction, and long-term memory, which combines intrinsic knowledge with external retrieval. Meanwhile, tools such as retrieval and planning modules empower agents to execute complex, goal-oriented tasks.

a) *Short-term Memory*: Multi-agent systems employ short-term memory to facilitate dynamic learning and adaptation. For example, AgentClinic [58] and MedChain [59] keep the full conversation and decision history for a single case. This allows the agent to adjust its questioning and diagnostic strategy over multiple turns based on past feedback, instead of relying only on a single-round input. Similarly, MEDDx-Agent [67] utilizes historical records and action memory to dynamically adjust their multi-step plans as new information becomes available. Furthermore, other systems focus on leveraging short-term memory to optimize multi-turn conversations and collaboration. MDTeamGPT [46] and EvoPatient [43] enhance a team’s reasoning by storing discussion history in a shared pool or dynamic libraries. Meanwhile, framework like PIORS [61] tracks dialogue history to provide patients with a coherent and personalized medical service. These multi-agent frameworks demonstrate that short-term memory is essential for an agent’s ability to dynamically learn and adapt during multi-turn clinical interactions.

b) *Long-term Memory*: In AI agents, long-term memory is achieved through two complementary paradigms, leveraging a model’s intrinsic knowledge and using retrieval-augmented generation (RAG). The MD2GPS [51] debate system perfectly illustrates this dual approach by combining a knowledge agent that accesses internal knowledge with a data agent that retrieves external data. Other frameworks also focus on directly enhancing intrinsic knowledge through training. For example, MedAgentGym [68] fine-tunes LLMs on real-world tasks. Similarly, LA-CDM [63] improve core skills and procedural knowledge through expert feedback and hybrid training paradigms, respectively. On the other hand, many frameworks extend memory via retrieval. For instance, AMDS [69] and MoodAngels [49] leverage external knowledge bases and databases for precise domain knowledge. Furthermore, hybrid models combine both of these paradigms. For example,

MedAssist [64] integrates web and EHR data. Recent work using o1 [65] as the main LLM for medical agents shows a more flexible form of long-term memory. As the agent handles different clinical cases, it can keep updating its reasoning steps, tool settings, and retrieved knowledge. In clinical AI agents, long-term memory is a hybrid system that combines intrinsic knowledge embedded during training with dynamic retrieval of external data to tackle complex clinical challenges.

To overcome the inherent knowledge limitations of LLMs, multi-agent systems employ data retrieval tools to access and analyze external information. A common strategy is Retrieval-Augmented Generation (RAG), adopted by systems such as TRIAGEAGENT [40] to ground triage and diagnosis in medical guidelines. Other frameworks, including ColaCare [39] and AIPatient [41], use specialized retrieval agents to access EHRs and knowledge graphs, thereby improving patient data analysis and medical question-answering. Specialized application include RareAgents [44] for rare disease diagnosis. MedAide [66] leverages RAG-based query rewriting and a Rotation Agent Collaboration (RAC) mechanism to perform intent-aware information fusion and coordinated multi-agent reasoning. Together, data retrieval and planning tools enable clinical AI agents to extend their knowledge, ground their reasoning in diverse evidence, and transform intricate clinical problems into structured, executable workflows.

4) *Use cases*: The practical value of clinical NLP agents is realized through their deployment in concrete clinical scenarios. Aligning with the agent-role taxonomy, these use cases bifurcate into those that directly engage and simulate the patient experience, and those that augment and streamline the work of healthcare providers. Patient Consultation and Clinical QA serve the former, while Clinical Documentation and Decision Support & Workflow Automation serve the latter. Together, they demonstrate a progression from isolated task-solving toward integrated, multi-agent frameworks that span the entire patient journey, from initial contact and diagnosis to treatment planning and record-keeping.

a) *Patient Consultation and Clinical QA*: This domain has undergone a significant evolution, transitioning from static information retrieval to dynamic, interactive reasoning systems. Early QA models such as KGAREVION [70] enhanced answers by retrieving and verifying information from knowledge graphs. The field then shifted toward reasoning-based and dialogue-driven approaches, where systems like MedAgents [16], AI Hospital [55], and MDAgents [38] employed collaborative, multi-round discussions to achieve consensus diagnoses. Extensions such as AIPatient [41] adapted these methods to patient-specific EHRs, while PIORS [61] uses multi-turn interactions in real outpatient visits, and DERA [60] uses agent dialogues to step-by-step refine conversation summaries and care plans. Parallel to these advances, simulation and benchmarking platforms like EvoPatient [43] introduced systematic evaluation by co-evolving patient–doctor dialogues. Overall, Patient Consultation and Clinical QA are progressing toward hybrid, multi-agent, and multimodal systems that not only retrieve and reason over medical knowledge but also simulate realistic clinical interactions, bridging the gap between information access and patient-centered decision support.

TABLE II
SUMMARY OF MEDICAL LLM AGENT FRAMEWORKS CATEGORIZED BY TASK TYPE, KNOWLEDGE HANDLING STRATEGIES, PIPELINE DESIGN, AND MODALITY.

Ref.	Task	Base LLM	Pipeline Type	Knowledge Handling	Tools	Modality	Public
[16]	Medical reasoning	GPT-4, GPT3.5	Multi-agent collaboration	Reasoning via role-playing	LLM unspecified	Text	✓
[38]	Medical decision support	GPT-4	Adaptive multi-agent planning	Shared medical knowledge coordination	Diagnostic tools	Text	✓
[39]	Electronic health record modeling	DeepSeek-V2.5	Multi-agent orchestration	Contextual EHR representation learning	EHR systems	Text	✓
[40]	Clinical triage	GPT-4, GPT-3.5-Turbo	Multi-agent collaboration	Guideline based medical knowledge	Clinical guideline tools	Text	✓
[55]	Multi-agent medical simulation	GPT-3.5, GPT-4, Wenxin 4.0, Qwen-Max	Environment based multi-agent system	Role based task reasoning with synthetic patients	Simulation framework	Text	✓
[48]	Multi-departmental diagnostics	GPT-4, GPT-3.5, Gemini-Pro-1.0, InternLM2-20B-Chat	Collaborative multi-agent reasoning	Real-world cross-department knowledge alignment	Clinical data systems	Text	✓
[54]	Differential diagnosis	GPT-4o, Llama3.1-70B, Llama3.1-8B	Modular agent framework	Explainable medical knowledge reasoning	Diagnostic modules	Text	✓
[44]	Rare disease consultation	GPT-4, Llama3.1-70B, Llama3.1-8B	Multi-disciplinary agent team	Rare disease knowledge integration	Medical databases	Text	✗
[46]	Team consultation simulation	gpt-4-turbo	Self-evolving multi-agent system	Continual knowledge adaptation across specialties	Consultation tools	Text	✓
[45]	Clinical dialogue	Qwen2.5-7B, DeepSeek-V3	Multi-agent reinforcement learning	Interactive clinical reasoning from feedback	Dialogue simulator	Text	✓
[56]	Synthetic patient-physician conversation	GPT-3.5, gpt-3.5-turbo-0613	Conditional dialogue generation	Knowledge conditioning via clinical notes	Conversation dataset	Text	✓
[58]	Adaptive medical reasoning	GPT-4, GPT-3.5	Autonomous agentic reasoning	Self-reflective adaptation to clinical context	Clinical reasoning tools	Text	✗
[59]	Sequential clinical benchmarking	gpt-4o-mini, claude-3.5-sonnet, llava-llama-3-8b-v1_1	Interactive multi-agent evaluation	Knowledge refinement through sequential tasks	Benchmark suite	Text	✓
[60]	Dialog-based reasoning	GPT-4	Multi-agent conversational reasoning	Contextual resolution across dialogue agents	Dialogue tools	Text	✓
[61]	Outpatient interaction simulation	GPT-4o	Multi-agent clinical workflow	Personalized patient knowledge modeling	Simulation tools	Text	✓
[42]	Patient simulator (online consultation)	Qwen2.5-72B-Instruct	SFT-based simulator	Strategy-guided from real dialogues	Dialogue simulator	Text	✓
[62]	Medical education copilots	GPT-3.5; GPT-4o-mini; Claude 3.5 Sonnet	Multi-agent copilot (student-patient-radiologist-expert)	Case-based learning with memory and ICD-10 retrieval	Radiology / ReportVQA tools	Text; image	✗
[47]	Medical diagnosis	Baichuan2-chat-13B	Hierarchical multi-agent	Simulates GP and specialists	Diagnostic tools	Text	✗
[49]	Psychiatry diagnosis	GPT-4o, DeepSeek-V3	Retrieval-augmented multi-agent	DSM-5 criteria and case-based retrieval	RAG, Symptom matching, Case retrieval	Text	✓
[50]	Mitigating cognitive bias	GPT-4 Turbo	Multi-agent conversation	Consensus via debate	Role-playing agents	Text	✗
[51]	Mendelian disease diagnosis	GPT-4.0	Multi-agent debate system	Internal knowledge + external data	Data agent, Knowledge agent	Text	✗
[52]	Personalized decision support	LLM-powered framework	Conversational agent	Coordinates external tools/data	External tools (Data, KB, AI Models)	Multimodal	✓
[53]	Medical QA	GPT-3.5	Agent-led discussions (RAG)	Multi-agent discussion for retrieval	Summarizer agent, Decision-maker agent	Text	✓
[57]	Psychiatrist/patient simulation	ChatGPT	Chatbot framework	Simulates dialogues via prompt design	Conversation tools, Prompt design	Text	✗
[63]	Clinical decision making	Llama-3-7B-Instruct	Hybrid training paradigm (RL)	Hypothesis-driven procedural reasoning	Diagnostic test requests (US, CT, Text Labs)	Text	✗
[64]	EHR comprehension	LLM unspecified	Hybrid model	Web and EHR data retrieval	RAG	Text	✗
[65]	Medical decision-making	o1-preview, GPT-4	Multi-agent framework evaluation	Enhanced reasoning (CoT, RAG)	CoD, MedAgents, AgentClinic	Text	✗
[66]	Complex clinical reasoning	GPT-4o, Claude-3.5-Sonnet, etc.	Multi-agent collaboration	Query decomposition	Rotation-based collaboration	Text	✗

b) Clinical Documentation: A central challenge in modern healthcare is the overwhelming burden of documentation. Clinical NLP agents are directly addressing this by automating the generation and management of medical records from unstructured data and dialogues. Moving beyond radiology

reporting, newer frameworks tackle a broader range of documentation tasks. The DERA framework [60] supports care plan generation and conversation summarization via a two-agent dialogue paradigm, while PIORS [61] enables interactive, multi-turn updating of outpatient records. A more advanced

approach is demonstrated by ColaCare [39], which simulates MDT-style discussions between a doctor and meta agents based on EHR predictions, and generates structured reports and a clear reasoning trail. Overall, advances in clinical documentation—from radiology reporting to dynamic patient record management—illustrate how multi-agent frameworks enhance both the efficiency and interpretability of clinical records.

c) *Decision Support and Workflow Automation*: This represents the most integrative frontier in clinical NLP agents, where the goal is to enhance both efficiency and quality of medical decisions by automating complex workflows. These systems reduce clinicians’ workload by automating routine tasks, generating clinical documentation, and offering evidence-based diagnostic suggestions. A major application is clinical documentation automation. DERA [60] focuses on long-form outputs such as care plans and conversation summaries. ColaCare [39] further extends this by simulating multi-agent discussions to produce consensus reports, bridging documentation with decision-making. Beyond documentation, automation also supports patient management and consultation flows. For instance, PIORS [61] automates outpatient reception and triage via multi-turn dialogue. On the decision support side, multi-agent systems like MedAgents [16] and AI Hospital [55] simulate collaborative doctor teams to refine diagnostic reasoning, and ColaCare applies a similar strategy to multi-disciplinary consultations. Overall, the field is evolving from isolated automation and knowledge-based systems toward integrated, multi-agent frameworks that unify documentation, patient management, and real-time decision support, driving both efficiency and diagnostic accuracy in clinical practice.

B. Medical Computer Vision

The perception module of a medical computer vision agent is tasked with interpreting a highly diverse and specialized array of imaging data. This extends beyond conventional radiological modalities such as CT, MRI, and X-Ray to encompass a broad spectrum of visual sources, including histopathology slides, endoscopic video, ophthalmological scans, and dermatological images. The complexity of this data demands agents that are not merely classifiers, but sophisticated systems capable of contextual reasoning and better aligning with clinical workflows. Tables III and IV provide a structured overview of key agents in this domain, detailing their applications and architectural approaches.

1) *Domain*: The core domain within the Medical Computer Vision category is medical imaging analysis. This area focuses on agents designed to process and interpret various forms of medical imagery, transforming raw visual data into structured, clinically relevant information. Although our taxonomy is organized by dominant data modality, many recent systems operate across multiple modalities. We therefore group frameworks under Medical Computer Vision whenever imaging provides the primary clinical evidence or serves as the central coordination signal for the agent workflow. In practice, this includes multimodal agents that combine imaging with text, signals, or structured clinical data while still centering their

planning and decision-making on visual findings. In this sense, modern “vision” agents increasingly act as hubs that fuse other modalities around imaging rather than remaining purely single-modality models.

a) *Medical imaging analysis*: Medical imaging analysis is undergoing a paradigm shift, driven by AI agents that move beyond passive perception toward active, reasoned interpretation. Recent agentic systems have advanced this field by combining reasoning, context-awareness, and guideline alignment. For example, MAGDA [71] and MedAgent-Pro [72] integrate clinical guidelines and structured workflows promote evidence-based, stepwise diagnostic reasoning aligned with clinical criteria. VoxelPrompt [73] expands vision-language capabilities, enabling interactive 3D imaging analysis and free-text, quantitatively grounded descriptions through LLM collaboration. Other works such as MedRAX [74] and PathFinder [75] demonstrate the power of multi-agent orchestration, unifying specialized tools for chest X-ray reasoning or gigapixel pathology slide analysis with transparent, interpretable outcomes. Beyond diagnosis, M3Builder [76] automates the end-to-end machine-learning pipeline for medical imaging, while SurgicalVLM-Agent [77] extends agentic support into image-guided surgery, acting as an interactive co-pilot for intraoperative planning and decision support. In summary, medical imaging agents are shifting from basic diagnostic support to more autonomous, clinically oriented systems that link image interpretation directly to practical clinical decisions.

2) *Agent roles*: Within medical computer vision, we focus on three agent roles that are most closely aligned with real-world clinical workflows and are actively being prototyped for clinical use: image-assisted diagnosis, medical image segmentation, and surgery support. Together, these agents are designed to turn raw images into structured findings, treatment plans, and intraoperative guidance, linking pixel-level analysis with the clinical decisions radiologists and surgeons ultimately make.

a) *Image-assisted Diagnosis Agent*: The Image-assisted Diagnosis Agent plays a central role in medical computer vision by leveraging imaging data to provide accurate, interpretable, and clinically relevant decision support. Recent advances showcase diverse approaches to this role. On one hand, models such as VoxelPrompt unify 3D imaging with language interaction to localize anatomy, quantify pathology, and enable automated radiological analysis [73]. Similarly, frameworks like MedRAX and CT-Agent extend these capabilities to chest X-rays and 3D CT scans, addressing complex spatial reasoning and aiming for more clinically reliable outputs [74], [78]. On the other hand, a growing body of work emphasizes multi-agent collaboration for report generation, where systems such as MoColl, Yi et al.’s framework, and other RRG-based agents combine domain-specific vision-language analysis with reasoning to synthesize coherent, trustworthy radiology reports [79], [80]. Expanding further, M3Builder [76] automates much of the end-to-end machine-learning pipeline for imaging, while MedAgent-Pro [72] uses a hierarchical, guideline-aware agent workflow to move from multi-modal perception to evidence-based diagnostic reasoning, aiming to make outputs more scalable and consistent with clinical standards. Reinforce-

ment learning-driven designs, as in MMedAgent-RL, push this paradigm toward adaptive collaboration between generalists and specialists, mimicking human-like clinical workflows [81]. Beyond technical accuracy, the Assertiveness-based BreastScreening-AI highlights the human-centered dimension of these agents, showing that communication style tailored to clinician expertise can reduce errors and increase trust without sacrificing diagnostic performance [82]. In sum, image-assisted diagnosis agents are evolving from isolated prediction models into collaborative, interpretable, and adaptive systems that bridge imaging data with trustworthy clinical decision-making.

b) Medical Image Segmentation Agent: Medical image segmentation agents provide the structural foundation of computer vision in medicine by accurately delineating organs and lesions across diverse imaging modalities. Serving as enablers for diagnosis, surgical planning, and treatment monitoring, they ensure that downstream agentic systems operate on precise and clinically meaningful visual information. FUAS-Agents [83] exemplify this role by combining MRI segmentation with dose prediction and surgical planning, enabling precise and personalized treatment strategies. This highlights how segmentation agents extend beyond image analysis to help optimize complex therapeutic workflows. MMedAgent [84] exemplifies this role by integrating and adapting state-of-the-art segmentation tools to medical contexts, enabling robust delineation across MRI, CT, and X-ray. By dynamically selecting the best tool for each task, it ensures precise and efficient segmentation that strengthens downstream clinical workflows. AgentPolyp [85] extends the role of medical image segmentation agents by combining semantic-guided enhancement with lightweight neural segmentation, ensuring accurate delineation of polyps under diverse degradation conditions. Its dynamic, agent-driven enhancement framework optimizes image quality for downstream analysis, demonstrating how segmentation agents can directly support robust and clinically meaningful workflows.

c) Surgery Support Agents: Surgery support agents are advancing beyond conventional medical imaging by combining real-time visual data with anatomical and contextual knowledge to improve precision and reduce risks during image-guided interventions. Recent works such as SurgicalVLM-Agent have shown how vision-language models can support interactive planning, instrument tracking, and surgical visual question answering, turning AI into an active intraoperative co-pilot [77]. Similarly, FUAS-Agents integrate MRI analysis, dose prediction, and treatment-planning rules to facilitate non-invasive ablation surgery, highlighting the potential of multimodal agents in personalized treatment planning [83]. Building on this, SurgRAW incorporates domain-specific priors with chain-of-thought reasoning and retrieval-augmented generation, moving from narrow, task-specific solutions toward more scalable surgical support [86]. Complementing these advances, Surgical-LLaVA embeds surgical images and videos into the language feature space, enabling multimodal dialogue and context-aware assistance with the potential to support decision-making in surgical scenarios. [87]. Together, these systems demonstrate a clear trend: AI

agents are evolving from specialized tools into holistic surgical copilots capable of supporting complex and dynamic clinical workflows. In summary, surgery support agents suggest a move from narrow imaging tools to more integrated, intelligent assistants that support surgical precision and personalization, but these systems still need strong clinical validation.

3) Tooling and memory: In medical computer vision, agents require specialized tooling and memory to handle the unique challenges of imaging data. Model architectures such as CNNs, vision transformers, and CLIP-like encoders extract rich spatial and semantic representations, while processing pipelines for denoising, registration, and enhancement aim to improve input reliability. Beyond perception, multimodal alignment links visual features with textual or temporal context, and memory mechanisms-including visual contextual memory and longitudinal tracking-enable agents to maintain consistency across sequential scans and surgical workflows. Together, these components allow vision-based agents to move beyond static interpretation toward dynamic, clinically adaptive support.

a) Model Architectures: As the foundation of medical computer vision agents, model architectures transform raw imaging data into spatial and semantic representations that support diverse clinical applications. Vision-language integration has proven especially effective: MAGDA leverages CLIP-based models within a multi-agent system to deliver guideline-aligned reasoning in radiology [71], while MedAgent-Pro combines vision-language architectures with retrieval-augmented reasoning to promote evidence-based diagnostic outputs [72], and RVQA shows how multi-agent vision-language frameworks improve accuracy, interpretability, and robustness in radiological question answering [88]. For segmentation tasks, MARL-MambaContour integrates reinforcement learning with a memory-enhanced backbone to capture long-range contour dependencies [89], and AgentPolyp demonstrates how CLIP-guided architectures with dynamic enhancement achieve robust analysis of noisy endoscopic images [85]. Extending beyond radiology, PathFinder illustrates how multi-agent architectures can emulate expert diagnostic workflows in pathology by coordinating triage, navigation, and diagnosis agents, resulting in accurate and explainable whole-slide image analysis [75]. Together, these advances highlight the growing role of architecture-level innovations in building trustworthy and clinically adaptive medical AI systems. In short, innovations in model architectures are driving a shift from narrow, task-specific designs to robust, explainable agents capable of supporting complex medical imaging workflows.

b) Image Processing Pipelines: The role of AI agents is evolving beyond simple task execution to shaping the entire medical imaging pipeline. At the preprocessing stage, systems like A-IDE adapt workflows to different anatomies, while AgentPolyp uses semantic understanding and reinforcement learning to build tailored enhancement pipelines for noisy or low-quality inputs [85], [90]. Moving further, frameworks such as M3Builder and VoxelPrompt show how agents can autonomously construct and adapt entire ML pipelines: M3Builder assembles preprocessing, training, and deployment

TABLE III
SUMMARY OF MEDICAL COMPUTER VISION-FOCUSED LLM AGENT FRAMEWORKS, CATEGORIZED BY TASKS, PIPELINES, KNOWLEDGE GROUNDING, TOOL USAGE, AND MODALITIES.

Ref.	Task	Base LLM	Pipeline Type	Knowledge Handling	Tools	Modality	Public
[91]	Multi-turn diagnostic reasoning virtual doctor	GPT-3.5, GPT-4, Llama-3	Multi-turn virtual doctor with multi-source RAG and concurrent disease checking	Multi-source retrieval over wearable sensor data, medical dialogues, textbooks, and guideline trees	Adaptive sensor-data retriever, semantic filter, mapping-based guideline-tree retriever	Text and wearable sensor data	✗
[73]	Three dimensional neuroimaging analysis	LLM unspecified	Agent orchestration	Image grounded visual reasoning without external retrieval	Imaging, three dimensional processing	Three dimensional MRI and CT	✗
[74]	Chest X-ray reasoning	GPT-4o, Claude-3.5-Sonnet	ReAct based orchestration	Tool grounded stepwise reasoning	Visual QA, segmentation	Chest X ray radiology	✓
[78]	3D CT QA and report generation	LLaVA-Med v1.5, DeepSeek-V3	LLM planner with anatomy-aware tool orchestration	Memory-guided few-shot retrieval, regional LoRA specialization	Anatomy LoRA tools, exemplar retrieval	3D chest CT + text	✗
[80]	Radiology report generation	GPT-4o, LLaVA-Med 1.5	Multimodal multi-agent workflow aligned to clinical reasoning	Retrieval-augmented; cross-modal grounding with structured intermediate outputs	CLIP-based retrieval	Chest X-ray images and radiology text	✗
[76]	Automated machine learning in medical imaging	Claude-3.7-Sonnet; GPT-4o; DeepSeek-V3; Gemini-2.0-Flash; Qwen-2.5-Max; Llama-3.3-70B	Multi-agent orchestration with auto-debugging	Workspace-grounded planning with datacards and code templates	File browsing, File read and write, Code execution	X-ray; CT, MRI	✗
[72]	Evidence-based multi-modal medical diagnosis	GPT-4o	Hierarchical agentic workflow	Guideline-grounded RAG with step reliability verification	visual models, coding tool	Fundus, 3D echocardiography, chest X-ray	✓
[81]	Multimodal medical VQA	Qwen2.5-VL	RL multi-agent	Specialist-opinion aggregation via curriculum RL	GRPO RL; curriculum learning; specialist LVLMS	Image+Text	✗
[82]	Medical imaging decision support	LLM unspecified	Interactive workflow	Model explanations with patient clinical arguments	Imaging viewer, segmentation highlights, BIRADS	Mammography; Ultrasound; MRI	✗
[83]	Treatment planning for FUAS	Qwen-3 14B	Multi-agent orchestration with RAG	record-grounded retrieval with memory and Reflexion	MedSAM-2; radiomics+XGBoost; vector-store RAG; calculator	3D MRI , text	✓
[84]	Multi-task medical agent	LLaVA-Med	Instruction-tuned planner with tool orchestration	Tool-output aggregation; retrieval augmented generation from Merck Manual	Grounding DINO; MedSAM; BiomedCLIP; ChatCAD	MRI, CT, X-ray, histology, gross; text	✓
[85]	Polyp segmentation	LLM unspecified	Reinforcement learning agent	CLIP guided semantics	CLIP, enhancement ops, segmenter	Endoscopy images	✗
[77]	Surgical co-pilot for pituitary surgery	Llama 3.2 Text, Llama 3.2 Vision	Interactive agent planning, parameter efficient fine tuning	Dataset grounded surgical workflow reasoning	MRI segmentation, video segmentation, overlay, instrument detection, VQA	MRI, endoscope, text	✗
[86]	Surgical scene understanding	GPT-4o	Hierarchical multi-agent CoT workflow	Retrieval-augmented reasoning with knowledge-graph checks	RAG, knowledge graph, panel evaluator	Multimodal	✓
[71]	Zero-shot chest X-ray diagnosis	Mixtral-8x7B Instruct (ablations: GPT-4, Llama-2-70B)	Multi-agent orchestration with CoT	Guideline-grounded reasoning with dynamic VLM prompting	CLIP (BioViL-T) tool use	X-ray	✗
[88]	Radiology visual question answering	Med-GEMMA; MMed-Llama-3-8B	Multi-agent orchestration	Retrieval-augmented reasoning with confidence-based validation	FAISS retrieval; reranker; validator	Chest X-ray + text	✗
[89]	Contour-based segmentation	LLM unspecified	Two-stage MARL with SAC	Contour prior memory	Imaging, reinforcement learning, Mamba policy	MRI, CT, endoscopy, microscopy	✗
[75]	Histopathology WSI diagnosis	Quilt LLaVA 7B, GPT 2	Multi-agent iterative workflow	Text conditioned navigation and patch evidence aggregation	Triage transformer, U Net navigator, VLM describer, LLM classifier	Pathology WSI	✓
[90]	Low dose CT denoising	GPT-4o	Single agent orchestration	Visual semantic cues from BiomedCLIP with prompt based expert selection	BiomedCLIP, RED CNN experts	X-ray CT	✗
[92]	Pathology differential diagnosis on whole slide images	PathChat Plus based on Qwen 2.5 14B	Multi-agent orchestration with hierarchical planning	Multi image visual and text reasoning with ROI grounded evidence	WSI viewer API OpenSlide; Trident preprocessing; reporting agent	Histopathology whole slide images	✗
[93]	Multimodal glaucoma diagnosis and reporting	GPT 4.1	Director coordinated Multi-agent	CAD grounded reasoning with optional clinician notes	SwinV2 classifier, SegFormer segmentor, role agents, director agent	Fundus image and text	✓
[94]	Alzheimer's disease analysis	GPT-4o	Hybrid planner	Image and text fusion with retrieval	Retrieval, imaging tools	Multimodal	✗

TABLE IV
CONTINUATION OF RECENT MEDICAL COMPUTER VISION-FOCUSED LLM AGENT FRAMEWORKS, EXTENDING TABLE III WITH ADDITIONAL TASKS, MODALITIES, AND PIPELINE DESIGNS.

Ref.	Task	Base LLM	Pipeline Type	Knowledge Handling	Tools	Modality	Public
[95]	Multimodal diagnosis via role collaboration	Medichat-Llama3-8B; HuatuoGPT-Vision-7B; Qwen-Audio-Chat; VideoLLaMA2-7B	Multi-agent orchestration	Retrieval augmented role-based synthesis	Web retrieval; agent tools; imaging tools	Text, image, audio, video	✓
[96]	Cardiology diagnosis and report	Llama 3.1 8B, LLaVA v1.5 13B	Multi-agent workflow, instruction tuning, guideline check	Guideline grounded, clinician adjudicated data	ECG image parser, table parser, guideline checker	Tabular, ECG images, text	✗
[97]	Pathology reporting	Gemini 2.5 Pro	Workflow orchestration	Case memory with instruction grounded reasoning	Slide viewer, region selection, image analysis, exemplar retrieval	Pathology	✗
[18]	Chest X-ray interpretation and report drafting	Phi-2 (2.7B) with SigLIP-Large encoder	Single-model VLM with instruction tuning	Parametric clinical knowledge (CheXinstruct), no retrieval	LLM unspecified	Chest X-ray and text	✗
[98]	Autonomous MRI reconstruction and artifact correction	GPT-4o; Gemini 2.0 Flash	VLM agent orchestration with consensus	VLM prior with confidence consensus; no external retrieval	CycleGAN reconstructor; motion corrector; denoiser	Brain MRI	✗

with multi-agent teams, whereas VoxelPrompt generates for complex 3D radiology tasks [73], [76]. During diagnostic reasoning, systems like PathFinder and SlideSeek [75], [92] mimic expert pathologists by iteratively exploring gigapixel slides to localize key evidence. Finally, for decision reliability, models such as MedChat orchestrate multiple role-specific agents to emulate a medical board, enhancing interpretability and mitigating errors through collaborative reasoning [93]. In summary, agents are shifting from passive executors to active orchestrators, dynamically automating, exploring, and coordinating across the full medical imaging lifecycle.

c) Multimodal Integration: Multimodal integration enables medical vision agents to combine imaging with complementary data such as clinical records, signals, or guideline knowledge. This fusion yields reasoning that is richer and more clinically grounded. For instance, systems like ADAgent and FUAS-Agents unify MRI/PET imagery with clinical profiles and treatment guidelines to support personalized diagnosis, prognosis, and surgical planning, thereby enhancing personalization and mitigating risk [83], [94]. Building on this idea, frameworks such as MAM, MedChat, and ZODIAC assign specialized roles (e.g., radiologist, general practitioner, cardiologist) within multi-agent teams to coordinate diverse modalities like text, images, audio, and ECG tracings, enabling transparent and role-aligned decision-making [93], [95], [96]. Similarly, CPathAgent and AgentPolyp demonstrate how integrating multi-scale imaging and text-driven guidance allows agents to navigate complex pathology slides or noisy endoscopic data with greater interpretability and robustness [85], [97]. Generalist systems such as MMedAgent and CT-Agent further illustrate how unifying MRI, CT, X-rays, and CXR tools under a flexible orchestration framework improves scalability, workflow efficiency, and reporting consistency without retraining [18], [78], [84]. Within the chest X-ray domain, CheXagent [74] plays a similar role as a generalist vision-language foundation model specialized for CXR interpretation. Beyond technical fusion, approaches like BreastScreening-AI and MMedAgent-RL highlight that effective multimodal integration also requires aligning outputs with human communication styles and dynamically coordinating specialist ex-

pertise, strengthening clinical trust and adaptability [81], [82]. Multimodal integration transforms isolated models into cohesive, clinically aligned agents, enabling more personalized, interpretable, and trustworthy decision support.

d) Visual Contextual Memory: Visual contextual memory empowers medical agents to retain and reuse critical cues across images, facilitating reasoning that transcends single-snapshot analysis. In pathology, for example, agents like PathFinder and CPathAgent actively build this memory by emulating a pathologist’s workflow; they dynamically navigate massive gigapixel slides to synthesize observations from scattered regions into a coherent diagnostic narrative [75], [97]. In radiology, this concept enables longitudinal tracking; VoxelPrompt, for instance, leverage attention mechanisms to link features across a sequence of 3D scans, supporting analysis and helping track changes over time, such as tumor growth [73]. In the dynamic intraoperative environment, agents like SurgicalVLM-Agent and SurgRAW create a shared, real-time memory of a procedure by collaboratively analyzing video feeds for consistent instrument tracking and workflow understanding [77], [86]. Beyond internal memory, some frameworks employ an external recall strategy, using retrieval agents to source and reference similar historical cases from a database, thereby grounding the current diagnosis in a rich context of clinical precedents [80]. Agents operationalize memory by dynamically exploring images, tracking changes over time, understanding real-time events, or retrieving historical cases to enable more robust, context-aware clinical reasoning.

4) Use cases: The practical value of medical computer vision agents is realized in key clinical applications that leverage their advanced roles, tooling, and memory. The following use cases including diagnosis and screening, image reconstruction and quality enhancement, and report generation demonstrate how agents translate foundational visual analysis into tangible clinical value, bridging the gap between algorithmic innovation and patient care. Across these applications, imaging serves as the primary source of clinical evidence, while structured covariates and textual records act as contextual signals that refine the agent’s reasoning.

a) Diagnosis and screening: In diagnosis and screening, agents have evolved from simple predictors into sophisticated systems that manage complex, multi-step reasoning. One primary strategy is task decomposition, where agents break down a diagnostic query into manageable steps. This is seen in systems like VoxelPrompt, which translates prompts into executable code for 3D analysis; MoColl, which uses a question-answer framework between a generalist and a specialist; and MedRAX, which orchestrates a suite of specialized tools to construct a transparent, verifiable reasoning chain [73], [74], [79]. Another advanced approach emulates the workflows of human experts. For example, PathFinder deploys a multi-agent team to collaboratively scan gigapixel slides like a pathologist, while MedAgent-Pro and MMedAgent-RL adopt hierarchical or reinforcement learning-based strategies to ensure diagnostic plans are both guideline-aligned and adaptively leverage specialist knowledge [72], [75], [81]. In practice, many of these diagnostic agents are inherently multimodal: frameworks such as ADAgent and FUAS-Agents treat MRI, CT, or PET images as the central evidence stream while integrating clinical covariates, laboratory values, and guideline knowledge as contextual inputs, reinforcing the view of medical computer vision agents as imaging-centered hubs rather than isolated single-modality models [83], [94]. These capabilities translate into direct clinical impact, with models like CheXagent streamlining report generation to significantly boost radiologist efficiency [18]. At a meta-level, frameworks such as M3Builder even automate the creation of these specialized diagnostic models, accelerating the entire development and deployment cycle [76]. In essence, diagnostic agents are transitioning from monolithic predictors into dynamic reasoners that deconstruct complexity, emulate expert workflows, and deliver measurable improvements in clinical efficiency and reliability.

b) Image Reconstruction and Quality Enhancement: In image reconstruction and quality enhancement, the agent acts as a dynamic decision-maker, transforming the process from a fixed pipeline into an adaptive, “diagnose-then-treat” paradigm. This agent-driven approach typically involves a two-stage process: first diagnosing the specific image degradation and then autonomously selecting the optimal tools to correct it. For instance, the AgentPolyp framework uses CLIP-driven semantic analysis to identify issues like “low-contrast” or “blur” in endoscopic images and then employs reinforcement learning to orchestrate a tailored sequence of enhancement operations [85]. Similarly, the AgentMRI system leverages a VLM to first identify the type of corruption in an MRI scan, such as motion artifacts or noise, before selecting the most suitable deep learning model to execute a precise correction workflow [98]. This agent-driven paradigm ensures that image enhancement is not a one-size-fits-all process but a context-aware, intelligent intervention.

c) Report Generation: In report generation, agents are moving beyond simple image captioning to act as clinical reasoners that synthesize complex information into actionable documents. A key strategy for enhancing reliability is multi-agent collaboration, where systems like ZODIAC and MedChat emulate a human medical team to fact-check findings, diversify clinical perspectives, and reduce the risk of hallucination [93], [96].

To ensure reports are grounded in evidence, other agents adopt procedural approaches. CPathAgent, for instance, mimics a pathologist’s workflow by dynamically navigating gigapixel slides, ensuring its conclusions are transparently linked to the visual evidence it examined [97]. This concept is extended by tool-using agents like VoxelPrompt, MedRAX, and MMedAgent, which decompose the task into verifiable steps—such as generating executable code or orchestrating specialized analysis tools—to construct a reliable final synthesis [73], [74], [84]. The impact of these agent-driven reports extends across the clinical workflow, with models like CheXagent improving radiologist efficiency by drafting preliminary reports, and systems such as FUAS-Agents and ADAgent pushing beyond diagnosis to generate complex, personalized therapeutic or management plans that combine imaging findings with patient-specific covariates and guideline knowledge [18], [83], [94]. Ultimately, agent-driven report generation leverages collaborative reasoning and procedural transparency to transform raw data into reliable clinical narratives and actionable treatment plans, enhancing both quality and efficiency.

C. Graph Learning

Graph Learning is a crucial domain for reasoning using structured knowledge. Table V compiles medical agent frameworks based on graph learning and their characteristics. Some frameworks (e.g., HEAL-KGGen, DrugAgent, KODA) simultaneously belong to the Graph Learning and Omics and Biology domains. We discuss their graph-centric aspects here and revisit their discovery-oriented roles in Sec. III-D.

1) Domain: The core focus within the Graph Learning domain is the utilization and construction of Biomedical and medical knowledge graphs. These structured representations are fundamental for enabling agents to perform complex reasoning tasks in medicine.

a) Biomedical and medical knowledge graphs: Biomedical and medical knowledge graphs (KGs) provide structured representations of biological and clinical entities, enabling AI agents to reason over complex biomedical information and support downstream decision-making. This domain is closely linked to Clinical Text Analysis, as its primary goal is to extract and structure knowledge from unstructured clinical text. Initial approaches focused on creating static knowledge bases from EHR data and clinical documents. For example, systems like AIPatient [41] and KG4Diagnosis [99] exemplify this foundational approach by constructing knowledge graphs from clinical text to enhance tasks like patient simulation and diagnosis. This method is further advanced by frameworks like KARMA from Lu et al. [100], which uses specialized agents to automate the continuous enrichment of knowledge graphs from unstructured text, performing tasks such as entity discovery and relation extraction. Recent studies show that coupling KGs with large language models (LLMs) enhances reasoning, grounds medical recommendations in verifiable knowledge, and reduces hallucinations [69], [101], [102]. For example, LLM-powered agents can translate natural language queries into graph operations to explore microbial interactions and accelerate drug target discovery [101], while others

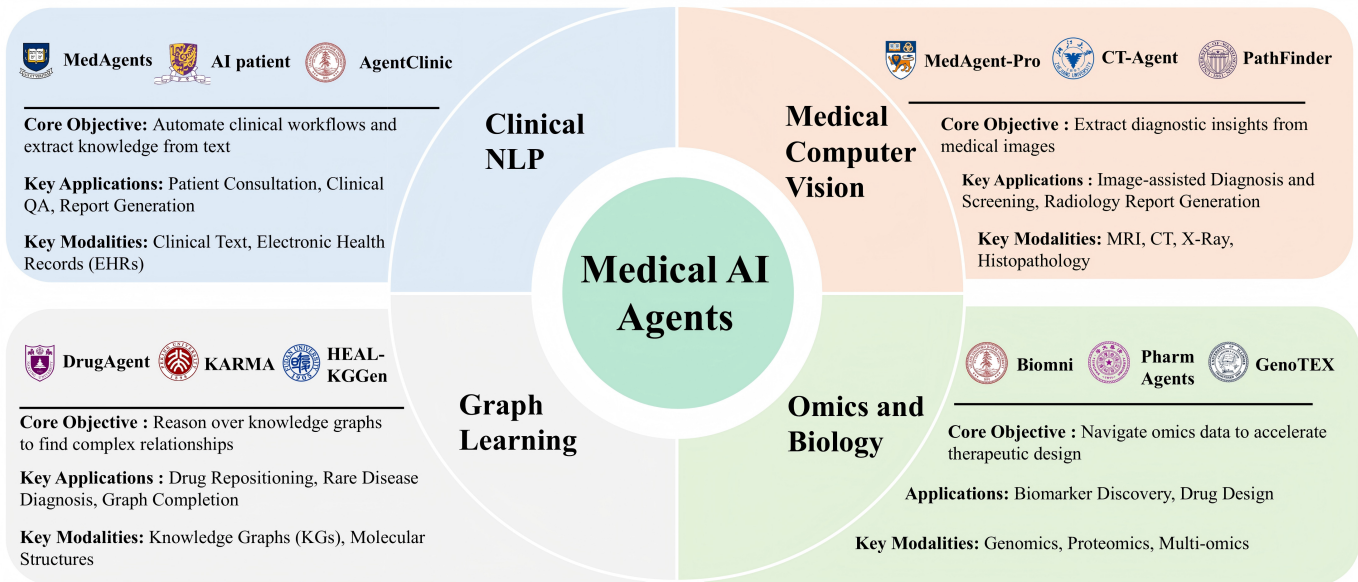


Fig. 4. A visual summary of the taxonomy for Medical AI Agents. Centered on the main theme, the figure branches into the four key domains proposed in this survey: Clinical NLP, Medical Computer Vision, Graph Learning, and Omics and Biology. Each quadrant details the domain’s Core Objective, Key Applications, and Key Modalities, and highlights representative agent models.

simulate cellular processes and refine hypotheses based on graph-grounded evidence [102]. Multi-agent systems further extend these capabilities: platforms such as PharmaSwarm and CLADD integrate curated KGs with LLM reasoning to generate, validate, and refine hypotheses in drug discovery, enhancing transparency and interpretability [103], [104]. Similarly, AIPatient links de-identified EHR-based KGs with agent systems for accurate medical QA and patient simulation, improving both educational fidelity and clinical trust [41]. Representative frameworks like ESCARGOT demonstrate how Cypher queries and retrieval-augmented generation (RAG) reduce hallucinations [105]. In parallel, systems such as KGAREVION [70], a KG-grounded QA agent that generates and verifies triplets against existing KGs, help reveal missing or conflicting knowledge. In summary, integrating biomedical knowledge graphs with LLM-based agents enhances reasoning, improves reliability, and makes complex medical knowledge more accessible.

2) *Agent roles:* In graph learning for biomedicine, AI agents take on complementary roles: discovering new relationships from fragmented data, completing and correcting knowledge graphs to ensure accuracy, and reasoning over structured representations for interpretable clinical insights. Through mechanisms such as cross-agent verification, hierarchical delegation, and adaptive orchestration, these agents transform biomedical knowledge graphs into dynamic, reliable, and clinically actionable resources.

a) *Relationship Discovery:* In graph learning for medicine, AI agents actively help discover relationships rather than just processing static data. Collaborative frameworks such as CLADD and KARMA organize agents into specialized teams, for example for KG construction, summarization, relation extraction, and schema alignment, and use cross agent verification and debate to improve reliability [100], [104].

Hierarchical multiagent systems such as KG4Diagnosis and HEAL-KGGen further resemble clinical teams. KG4Diagnosis uses a general practitioner agent to route cases to specialist agents that work over an automatically constructed diagnostic KG covering hundreds of common diseases, narrowing differential diagnoses through graph based decision paths [99]. HEAL-KGGen instead combines multiomics data with an automatically constructed medical KG and specialized agents to identify diagnostic relationships and new genetic biomarkers across disease cohorts [106]. Likewise, frameworks such as DrugAgent integrate evidence from knowledge graphs, machine learning models and literature mining to give interpretable predictions of drug target interactions and drug repurposing candidates [10]. In summary, AI agents in graph learning help discover, organize and check biomedical relationships, turning fragmented evidence into usable knowledge for medicine.

b) *Graph Completion:* Graph completion in biomedicine positions AI agents as active builders of missing knowledge, where they not only infer unobserved links but also validate and correct errors to ensure trustworthy insights. Recent advances such as BioKGBench [107] and its BKGAgent baseline explicitly target KG error discovery (KGCheck) rather than downstream clinical prediction, so we categorize them under graph completion / KG quality control. Multi-agent frameworks like KARMA [100] extend this role by assigning specialized agents to tasks such as entity discovery, relation extraction, schema alignment, and conflict resolution, achieving both precision and scalability through cross-agent verification and adaptive prompting. Similarly, systems like ClinicalAgent [108] highlight how graph completion supports clinical trial intelligence by retrieving drug data, analyzing diseases, and predicting trial outcomes, thereby filling critical knowledge gaps and guiding trial design. In summary, graph completion

enables agents to evolve biomedical knowledge graphs into accurate, dynamic, and clinically actionable resources.

c) Graph-based Reasoning: Graph-based reasoning agents harness biomedical knowledge graphs (KGs) as structured priors to deliver accurate and interpretable insights for clinical use. By linking heterogeneous data sources and explicitly reasoning over graph paths and subgraphs, they can explain predictions through relationships such as gene–disease–drug and protein–drug interactions. Multi-agent designs further enhance this process by assigning specialized agents to diagnosis, patient–trial matching, and knowledge verification, and coordinating their inferences via the underlying KG. For example, DrugAgent combines KG reasoning with external search to provide transparent treatment suggestions [10], KG4Diagnosis uses hierarchical agents over a KG to narrow down disease hypotheses [99], and MAKAR improves trial matching by reasoning over enriched eligibility criteria and patient profiles [109]. Beyond prediction, frameworks such as AIPatient extend these roles to clinical dialogue, grounding conversational recommendations in KG-derived evidence [41]. Collectively, these systems turn static KGs into dynamic substrates for biomedical reasoning and trustworthy decision support, ensuring that graph-based agents remain both clinically useful and explainable.

3) Tooling and Memory: Graph-centric medical agents rely on a combination of long-term structured memory and flexible retrieval mechanisms to support reliable reasoning. In this domain, biomedical knowledge graphs serve as the primary memory substrate, while LLM–RAG pipelines, query planners, and graph-centric toolchains (e.g., Cypher executors, ontology mappers, and visualization front-ends) enable agents to traverse, update, and operationalize these graphs. Together, these components transform static KGs into dynamic, queryable memory that can be aligned with clinical dialogue, discovery workflows, and decision support.

a) LLM–RAG Combination: The integration of LLMs with retrieval-augmented generation (RAG) has become a powerful paradigm for enhancing reasoning and graph-based inference in biomedical applications. Several systems demonstrate this synergy. For medical dialogue, AMDS integrates medical knowledge graphs and process knowledge with LLM-based agents, enabling more grounded and interpretable consultations [69], while AIPatient combines knowledge graphs with reasoning-augmented RAG to provide highly accurate and robust simulated patient interactions, demonstrating strong potential for medical education and training [41]. For biomedical discovery, KODA links LLM-driven agents with a Neo4j microbiome knowledge graph to reveal key microbial genes as drug targets through natural language queries [101]. To ensure adaptability and reliability, AGENTiGraph supports natural language interaction with adaptive knowledge graphs for accurate task execution [110], while MedAide enhances multi-agent reasoning by introducing syntactic-regularized query decomposition and adaptive intent matching [66]. Furthermore, KGAREVION complements RAG-style systems by prompting LLMs to generate candidate triplets and then verifying them against biomedical KGs, filtering out inconsistent facts and thereby improving robustness on complex medical QA tasks

[70]. In summary, LLM–RAG integration helps turn biomedical knowledge graphs into reliable, explainable, and adaptive tools for discovery, dialogue, and decision support.

b) Graph-enhanced Evidence-based Reasoning: Graph-enhanced evidence-based reasoning uses structured relational knowledge to gather and explain mechanistic evidence, rather than relying only on correlation-based predictions. In the DrugAgent system [10], this role is handled by a KG Agent and a Reasoning Agent that work over biomedical knowledge graphs to compute interaction scores and retrieve path-level evidence. By combining graph-derived mechanisms with machine learning predictions and literature signals, DrugAgent gives step-by-step, path-based explanations for each drug–target interaction or repurposing suggestion. HEAL-KGGen [106] extends this idea by turning a hierarchical, multi-source medical KG into contextual prompts for specialized agents, guiding LLMs to rank and evaluate genetic biomarkers along KG-based pathways and reducing hallucinations in benchmark diagnostic tasks. In the KG4Diagnosis framework [99], decision relationships are encoded in a diagnostic KG, allowing a hierarchical multiagent system to reason step by step about differential diagnoses (and in principle treatment options), with the KG acting as a structural prior that constrains LLM reasoning and helps reduce hallucinations. Together, these systems show how graph-enhanced agents can use biomedical KGs as working evidence bases for interpretable, evidence-based reasoning, rather than as static collections of associations.

4) Use cases: Building on the above roles and tooling, graph-learning-based agents are being deployed in several high-impact biomedical scenarios. Representative applications include drug repositioning, clinical trial outcome prediction and patient matching, and multi-omics–driven diagnosis, where agents combine KG reasoning with neural predictors and external tools to generate interpretable, hypothesis-driven insights. These use cases illustrate how structured graph representations and agentic orchestration jointly translate complex biomedical knowledge into actionable clinical and translational decisions.

a) Drug Repositioning: The convergence of AI agents, graph learning, and drug repositioning is triggering a profound paradigm shift. By constructing and analyzing heterogeneous biomedical knowledge graphs, graph learning enables AI to evolve from simple feature-based predictions to a deep understanding of complex biological mechanisms. AI agents, in turn, transform this capability into an autonomous, efficient, and collaborative drug discovery workflow, transitioning from passive tools to proactive research assistants. This is exemplified by agentic RAG frameworks such as CLADD [111], which overcome the need for costly fine-tuning by dynamically retrieving evidence from diverse biomedical knowledge bases. Complementing these automated pipelines, interactive platforms like Talk2KnowledgeGraphs and AGENTiGraph translate natural-language queries into graph traversals over disease, drug, and target knowledge graphs, allowing researchers to rapidly surface repositioning hypotheses and inspect the supporting evidence paths in an explainable manner [110], [112]. Together, these graph-centric agents help turn drug repositioning into a continuous, hypothesis-driven process that

is both scalable and transparent.

D. Omics and Biology

Finally, we analyze agents in the Omics and Biology domain, which hold significant potential for accelerating biomedical discovery. Table VI summarizes the cutting-edge research in this field.

1) *Domain*: This section examines the primary domains where Omics and Biology agents operate. We focus on Genomics and Proteomics, where agents interpret sequence data and molecular interactions, and Multi-omics Data Analysis, where they integrate information across different molecular layers to gain holistic biological insights.

a) *Genomics and Proteomics*: AI agents are pioneering the next frontier in medicine by autonomously navigating the complex, high-dimensional landscape of 'omics data to uncover the molecular foundations of disease. This is achieved through several key advancements. First, agents establish a structured understanding of the biological world by dynamically constructing knowledge graphs from raw multi-omics data, as seen in hierarchical systems like HEAL-KGGen, which mirrors clinical workflows to ensure reliable reasoning [106]. Building on this knowledge, generalist agents like Biomni learn to act by systematically mining scientific literature to create a unified action space of tools and protocols, enabling them to autonomously compose and execute novel discovery workflows [113]. Moving beyond mere workflow execution, these systems now function as autonomous scientists. For example, BioDiscoveryAgent actively participates in the research cycle by designing its own experiments and reasoning about the outcomes to navigate the vast hypothesis space efficiently [114]. This capability is powered by specialized foundation models like TourSynbio-7B, which are trained to natively understand the language of protein sequences [115]. Leveraging such models, agentic frameworks can then orchestrate complex bio-design tools, translating high-level goals into multi-step protein engineering workflows that are validated in real-world labs. AI agents in omics are rapidly evolving from data analysis tools into autonomous scientific partners that can build knowledge, orchestrate complex workflows, and drive the entire discovery cycle from hypothesis generation to experimental validation.

b) *Multi-omics Data Analysis*: Decoding the intricate interplay across genomic, proteomic, and other molecular layers is a central challenge in biology, a task for which AI agents are uniquely equipped. To ensure the reliability of their findings, agents like GeneAgent [116] are endowed with a self-verification capability, compelling them to autonomously query expert-curated biological databases. This process allows them to fact-check their own interpretations of gene sets from genomic or proteomic data, significantly reducing hallucinations and enhancing the trustworthiness of the final biological insight.

2) *Agent roles*: In the Omics and Biology domain, AI agents typically assume two complementary roles: end-to-end Drug Discovery Assistants that orchestrate multi-stage pipelines from target identification to lead optimization, and

Target Screening Agents that focus on systematically prioritizing disease-relevant genes and pathways. Together, these roles translate raw omics data and biological prior knowledge into concrete therapeutic hypotheses and experimentally testable candidates.

a) *Drug Discovery Assistant*: In the role of a Drug Discovery Assistant, an AI agent acts as an intelligent orchestrator that streamlines the entire therapeutic development workflow. This orchestration is increasingly modeled after human research teams, where a principal investigator (PI) agent guides interdisciplinary specialists to solve complex problems and translate computational work into experimentally validated discoveries [117]. These multi-agent ecosystems decompose the entire discovery pipeline into distinct stages—from target identification to preclinical analysis—with specialized agents collaborating in a structured workflow to automate the end-to-end process [118], [119]. The effectiveness of these teams is powered by two key innovations. First, specialized foundation models are trained to natively understand the complex language of molecules, such as proteins, providing deep domain knowledge [115]. Second, novel architectures grant agents a form of self-awareness, allowing them to dynamically reconfigure their internal structure to access the right knowledge for a specific molecular design or property prediction task [120]. Finally, to ensure this complex orchestration is rational and not aimless, advanced frameworks like PiFlow guide the agent's decision-making by treating discovery as a structured uncertainty reduction problem, ensuring a systematic and accelerated path to identifying high-potential candidates [121]. The modern Drug Discovery Assistant agent acts as a sophisticated orchestrator, leveraging team-based architectures, specialized foundation models, and rational guidance frameworks to automate the entire research pipeline from hypothesis to validation.

b) *Target Screening Agent*: The Target Screening Agent acts as a specialized data detective, synthesizing evidence from genomics, proteomics, and pathway analysis to pinpoint and rank the most viable targets for therapeutic intervention. This detective work is being implemented through several increasingly sophisticated approaches. First, to ensure systematic rigor, agents like GenoAgent autonomously execute entire computational genomics pipelines, methodically identifying disease-associated genes through standardized data analysis workflows [122]. A second approach focuses on interactive discovery within a specific biological niche; frameworks such as KODA leverage specialized knowledge graphs for the gut microbiome, translating researchers' natural language questions into precise queries to find novel drug targets [101]. Ultimately, this specialized role is being operationalized within end-to-end multi-agent ecosystems like PharmAgents. In these systems, a dedicated target discovery agent identifies and validates targets before seamlessly handing them off to downstream agents, fully automating the workflow from initial discovery to preclinical analysis and completing the transition from a standalone tool to an integrated component of a larger discovery engine [118]. The Candidate Target Screening Agent automates target discovery through three primary approaches: executing systematic pipelines, enabling

TABLE V
SUMMARY OF RECENT GRAPH LEARNING-BASED MEDICAL LLM AGENT FRAMEWORKS, COVERING THEIR TASKS, KNOWLEDGE HANDLING STRATEGIES, TOOLS, AND SUPPORTED MODALITIES.

Ref.	Task	Base LLM	Pipeline Type	Knowledge Handling	Tools	Modality	Public
[69]	Adaptive medical dialogue	LLM unspecified	Multi-agent system (Manager, Retriever, Refiner)	Integrates medical KGs and process knowledge	Medical KGs, retrieval tools	Text, Graph	✗
[101]	Microbiome drug target discovery	GPT-4o	Multi-agent orchestration	Schema-guided knowledge-graph reasoning and retrieval	Neo4j KG, Cypher executor, web search, LLM reviewers	Text; graph	✓
[102]	Survey of biomedical AI agents	LLM unspecified	Perspective and taxonomy	Literature-grounded synthesis on agentic use of KGs	LLM unspecified	Text	✗
[103]	Hypothesis-driven drug discovery	GPT-4, Gemini 2.5 Pro, TxGemma	Multi-agent orchestration	Graph-grounded retrieval with memory and simulation feedback	Knowledgegraph,pathwayenrichment,PETSSimulation,iBAMbinding,literaturemining	Text; omics; molecules	✗
[104]	Drug discovery question answering	GPT-4o mini	Multi-agent orchestration	KG-based retrieval with planning and molecular annotations	Knowledge graph, annotation database, captioning model, similarity search	SMILES; text	✗
[41]	Medical diagnosis with knowledge graph	LLM unspecified; BioBERT for extraction	Hierarchical multi-agent	KG construction and augmentation with expert validation	Entityandrelationextraction,Neo4j knowledgegraph,multi-agentsystem	Text; graph	✓
[105]	KG-grounded biomedical reasoning	GPT-4o mini (benchmark; model-agnostic)	Dynamic Graph-of-Thoughts agent with code execution	Cypher over KGs with RAG fallback	KGdatabase;Cypher;vectordatabase;Pythonexecutor	Text; graph	✓
[100]	KG enrichment from scientific literature	GLM-4; GPT-4o; DeepSeek-V3	Multi-agent orchestration with hierarchical controller	Summarization;entity/relationextraction;schemalignment;conflictresolution;LLMverification	KGtools;ontologymapping;retrieval;evaluationagents	Text	✓
[70]	Medical QA with KG revision	LLaMA 3 8B; LLaMA 3.1 8B	Single-agent action loop	KG-grounded triplet generation and verification (UMLS mapping)	Medical KG; UMLS; structural embeddings	Text	✓
[110]	KG interaction and QA platform	GPT-4o; Gemini 1.5 Pro; LLaMA 3.1	Multi-agent orchestration	KG traversal and Cypher reasoning with dynamic KG integration	Neo4j, Cypher, NER, relation extraction, visualization	Text; graph	✗
[99]	Medical diagnosis reasoning with hierarchical agents	GPT series; BioBERT	Hierarchical multi-agent with GP and specialists	Automated KG construction; semantic entity/relation extraction; human-guided reasoning	BioBERT;Neo4j;SNOMED-CT;UMLS;PyTorch	Text; structured KG	✗
[106]	Medical diagnosis reasoning from multi-omics	GPT; BioBERT	Hierarchical multi-agent with GP and specialists	KG construction from multi-omics; semantic entity/relation extraction	BioBERT;Neo4j;SNOMED-CT;UMLS;PyTorch	Text; structured KG; omics	✗
[10]	Drug-target interaction prediction	GPT-4o; GPT-4o mini	Multi-agent coordination with specialized agents	Integration of ML prediction, biomedical KGs, and web evidence	DeepPurpose;AutoGen;BingSearchAPI;biomedicalKGs	Text; molecules; protein; graph	✓
[108]	Clinical trial outcome prediction and reasoning	GPT-4	Multi-agent system with planning and reasoning agents	IntegrationofDrugBank,Hetionet,ClinicalTrials.gov,andLLM-generateddata	Enrollmentmodel;drug/diseaseriskmodels;databaseretrievers;ReAct;LLMmodules	Text; structured data; molecules	✓
[109]	Clinical trial patient matching and eligibility reasoning	GPT-4o; Qwen; LLaMA; Gemma; Mistral	Multi-agent system with augmentation and reasoning modules	Criterion enrichment via retrieval, web search, self-augmentation; structured reasoning over EHR	Routeragent;knowledgeaugmentation;critic;reasoningagent;matchingagent	Text; structured EHR	✗
[112]	Interactive biomedical KG exploration	LLM unspecified	Interactive Graph-RAG platform	Natural-language-to-graph traversal and subgraph retrieval	Biomedical knowledge graphs, query engine	Text; graph	✗

interactive exploration of specialized knowledge graphs, and operating as a key module within larger, end-to-end discovery ecosystems.

3) *Tooling and Memory*: In the Omics and Biology domain, agent capabilities are largely defined by two complementary pillars: powerful computational engines for modeling molecules, and structured memories for grounding biological reasoning. On the one hand, protein foundation models and molecular simulations enable agents to reason over sequences, structures, and dynamics at scale; on the other hand, biomedical knowledge graphs and graph-based RAG provide persistent, verifiable context that links these computations back to prior evidence and expert knowledge.

a) *Protein FMs and Simulation Tools*: Protein foundation models (FMs) and molecular simulations serve as the core computational engines that power an AI agent’s ability to reason about protein structure, function, and dynamics. A

key development is the creation of multimodal FMs like ProteinGPT, which fuse knowledge from specialized protein sequence and structure encoders to enable comprehensive, conversational analysis of target proteins [123]. These engines are evolving further with models such as TourSynbio-7B, which are trained to natively understand protein sequences as a new language, providing deeper and more efficient domain knowledge [115]. Agentic platforms then integrate these powerful engines into collaborative workflows. Systems like ProtAgents and the ‘Virtual Lab’ assign specialized agents to manage different tools—from generative models to physics-based simulations—and orchestrate them into novel discovery pipelines [117], [124]. This synergistic approach of combining diverse computational engines has proven highly effective, leading to the design of new therapeutics and experimentally validated scientific breakthroughs. AI agents leverage advanced protein foundation models and simulations as core

engines, orchestrating them within multi-agent workflows to translate computational design into experimentally validated scientific discoveries.

b) Biomedical knowledge graphs and Graph RAG:

Biomedical knowledge graphs (Bio-KGs) serve as the structured long-term memory for AI agents, which they access via Graph RAG to ensure their reasoning is grounded in verifiable facts. This process begins with the memory’s creation and maintenance. Frameworks such as HEAL-KGGen take this further by dynamically constructing KGs from complex multi-omics data, transforming retrieved subgraphs into contextual prompts that directly guide and constrain the LLM’s reasoning process [106]. PerTurboAgent [125] utilizes historical records and action memory to dynamically adjust their multi-step plans as new information becomes available. Within multi-agent discovery workflows, this memory access is often operationalized through specialized roles. Systems like CLADD and DrugAgent deploy dedicated ‘Knowledge Graph Teams’ or ‘KG Agents’ responsible for querying the graph and retrieving structured evidence to support the team’s overall hypothesis [10], [104]. Ultimately, this synergy democratizes access to complex information; interactive platforms like AGENTi-Graph explicitly use Graph RAG to translate natural language questions into graph traversals, providing explainable, fact-grounded answers for scientific exploration [110].

4) Use cases: Building on these roles and toolchains, Omics and Biology agents are now being deployed in concrete, high-impact biomedical workflows. In this survey, we highlight two representative application areas: biomarker discovery, where agents mine high-dimensional omics data to uncover disease signatures, and therapeutic design and prediction, where they explore chemical and protein spaces to propose and refine candidate treatments.

a) Biomarker discovery: As a core use case in precision medicine, AI agents act as automated discovery engines, systematically mining high-dimensional omics data to identify novel biomarkers for clinical applications. To achieve this, systems like GenoAgent are designed to autonomously execute entire computational genomics pipelines, methodically analyzing data to validate disease-associated genes as potential therapeutic targets [122]. More advanced frameworks such as HEAL-KGGen employ a hierarchical multi-agent system that not only analyzes the data but also dynamically constructs its own knowledge graph from multi-omics inputs, enabling deeper, context-aware reasoning for biomarker validation [106]. This specialized function is now being integrated into end-to-end discovery ecosystems like PharmaAgents, where a dedicated target discovery agent identifies and validates biomarkers before handing them off to downstream agents, fully automating the pipeline from initial discovery to preclinical analysis [118].

b) Therapeutic Design and Prediction: In the realm of Therapeutic Design and Prediction, AI agents navigate the vast chemical space to both generate novel drug candidates and iteratively refine existing molecules for improved efficacy and safety. This process often begins with understanding the biological target, where agentic frameworks like the ‘Virtual Lab’ orchestrate tools such as ESM and AlphaFold to deter-

mine protein structures, laying the foundation for structure-based design [117]. This iterative refinement is supported by specialized systems at every step. A critical task is accurately predicting the drug-target interaction, for which multi-agent systems like DrugAgent synthesize evidence from diverse sources to produce a single, explainable binding prediction [10].

IV. CHALLENGES

Despite rapid progress, the deployment of medical AI agents in real-world clinical settings is constrained by several fundamental challenges. These obstacles span technical, clinical, and regulatory dimensions and must be systematically addressed to achieve safe and effective integration into healthcare. This section highlights three core challenge areas: (1) limitations in medical data quality and availability, (2) shortcomings in clinical modeling and reasoning, and (3) barriers to clinical deployment and adoption.

1) Medical Data Quality and Availability: A primary obstacle is the pervasive fragmentation and heterogeneity of clinical data. Healthcare ecosystems are siloed, with hospitals, insurers, and research institutes maintaining independent electronic health record (EHR) systems, imaging archives, and genomic repositories.

Even with standards like HL7 and FHIR [126], inconsistent implementation hinders interoperability, preventing the construction of unified patient records. This fragmentation directly leads to incomplete clinical representations and biased agent reasoning.

The challenge intensifies with multimodal integration. Clinical narratives are unstructured and noisy, radiological images vary in resolution and annotation quality, laboratory results follow yet another structure, and omics profiles are both high dimensional and sparse [127]. Without reliable methods of alignment, these modalities remain isolated. Small sample sizes further exacerbate the challenge. Rare diseases provide too few examples to train robust multimodal models, which means that AI agents struggle to generalize across different patient populations. Current few-shot and fusion techniques provide partial solutions, but they fail to capture the depth of domain-specific semantics [128]. This underscores a critical bottleneck: the lack of scalable, reliable pipelines for multimodal integration.

Compounding these issues, clinical datasets are prone to quality issues. Human annotation errors, ambiguous diagnostic guidelines, and incomplete documentation introduce inconsistencies that propagate through downstream reasoning. This propagation produces distortions in predictions and weakens the reliability of clinical decisions. Beyond annotation noise, biomedical knowledge itself is dynamic. Drug safety alerts, diagnostic classifications, and treatment guidelines evolve continuously. Static corpora risk embedding outdated practices, and agents trained on them can unintentionally reproduce obsolete knowledge. Retrieval-based augmentation and continual learning pipelines are promising, but their maturity is still far from meeting the strict requirements of clinical practice.

TABLE VI
SUMMARY OF RECENT OMICS AND BIOLOGY-ORIENTED LLM AGENT FRAMEWORKS, COVERING THEIR TASKS, KNOWLEDGE HANDLING STRATEGIES, TOOLS, AND SUPPORTED MODALITIES

Ref.	Task	Base LLM	Pipeline Type	Knowledge Handling	Tools	Modality	Public
[106]	Medical diagnosis reasoning	GPT; BioBERT	Hierarchical multi-agent with GP and specialist LLMs	Automated knowledge graph construction; semantic entity and relation extraction; human-guided reasoning	BioBERT; Neo4j; SNOMED-CT; UMLS; PyTorch	Text; structured medical KG	✗
[113]	General-purpose biomedical research automation	GPT-4o	Code-centric agentic system	Retrieval-augmented planning and unified action space	150 biomedical tools; 105 software; 59 databases	Text; multi-omics; clinical; sensor data	✓
[114]	Genetic perturbation experiment design	Claude 3.5 Sonnet; GPT-4o; Claude 3 Haiku	Closed-loop biological experiment planning	Iterative reasoning with literature, database, and observation feedback	PubMed search; Reactome pathway gene query; AI critic	Genomic data; phenotype response	✓
[115]	Decision Support	InternLM2-Chat-7B	(—)	Pretrained protein/structure priors	ProtFM; simulation	Protein sequence/structure	✓
[116]	Gene Set Knowledge Discovery	GPT-4	Self-verification workflow	Self-verification against domain databases	Profiler, Enrichr, E-utils, Custom APIs	Text, Genomics/Proteomics Data	✗
[117]	Autonomous Scientific Discovery	GPT-4, Llama 2	Multi-agent orchestration	Episodic memory	self-reflection	Scientific simulation tools, ProtFM, Protein, Chemical, Materials	✓
[118]	End-to-end Drug Discovery Automation	GPT-4o, DeepSeek-V3/R1	Multi-agent orchestration	Experience-based self-evolution	UAlign, MetaTrans, DrugCLIP, DecompDiff	Text, Molecular Structures, Omics	✗
[119]	End-to-end Drug Discovery	GPT-4o, Llama 3.1, Gemini 1.5	Multi-agent orchestration	Task decomposition and validation	Generative models, Docking, RDKit, Property prediction	Text, Molecular Structures	✗
[120]	Autonomous Molecular Design	GPT-4	Self-reconfiguring multi-agent system	Principle-driven reasoning with contextual adaptation	ProtFM, GNNs, Molecular Simulation Tools	Text, Molecular Structures	✓
[121]	Principle-Aware Scientific Discovery	GPT-4	Uncertainty-aware agent workflow	Principle-driven reasoning	Retrieval, Experiment Simulators	Text, Scientific Data	✗
[122]	Genomics Task Execution Benchmarking	GPT-4, Llama-2-70B, GPT-3.5-Turbo	Agentic workflow for evaluation	Episodic task memory for multi-step tasks	Genome browsers, Genomic analysis tools	Text, Genomic Data	✓
[101]	Microbiome Drug Target Discovery	GPT-4o	Multi-agent orchestration with KG querying	Schema-guided knowledge-graph reasoning and retrieval	Neo4j KG, Cypher executor, Web search	Text, Graph Data	✓
[123]	Protein Property Prediction	Llama 3, Vicuna, Mistral	Multimodal fusion with instruction tuning	Sequence and structure alignment with text	ESM-2, GVP-GNN, AlphaFold 2	Protein Sequence Structure	✓
[124]	Autonomous Protein Discovery	GPT-4	Multi-agent orchestration	Episodic memory with self-reflection	AlphaFold, ESM-2, Molecular Dynamics Simulators	Protein Sequence, Structure	✓
[104]	RAG-based QA for Drug Discovery	GPT-4o mini	Multi-agent collaboration	RAG over molecular knowledge with anchor drug retrieval	Knowledge Graphs (PrimeKG), GNNs, Annotation DBs (PubChem)	Molecular Data	✗
[10]	Drug-Target Interaction Prediction	GPT-4o, GPT-4o mini, o3-mini	Multi-agent coordination with CoT/ReAct reasoning	Integration of evidence from ML, KG, and web search	DeepPurpose, AutoGen, Bing Search API, Biomedical KGs	Text, Molecular Data, Graph Data	✓
[68]	Code-centric biomedical reasoning	29 LLMs (OSS Proprietary)	Agentic training environment	Interactive feedback, Verifiable ground truth	Code interpreter, Docker sandbox, Biomedical libraries	Text, Code, Tabular, Genomics	✓

2) *Clinical Modeling and Reasoning*: One of the most critical limitations of current medical AI agents is the lack of interpretability [129]. Unlike traditional clinical decision support systems [130], which often follow explicit rules and can be audited step by step, LLM-based agents often function as “black boxes” with opaque internal reasoning. Clinicians who rely on such systems are often unable to trace how a specific recommendation was generated. This opacity reduces trust and makes it difficult to incorporate AI advice into

clinical workflows where accountability is essential [131].

The problem becomes more serious in high stakes environments such as oncology or critical care, where decisions involve complex interdependencies among drug interactions, comorbidities, and long-term prognostic factors. Without transparent reasoning pathways, clinicians cannot verify whether the agent has properly considered these variables, increasing the risk that spurious correlations or latent biases influence recommendations undetected.

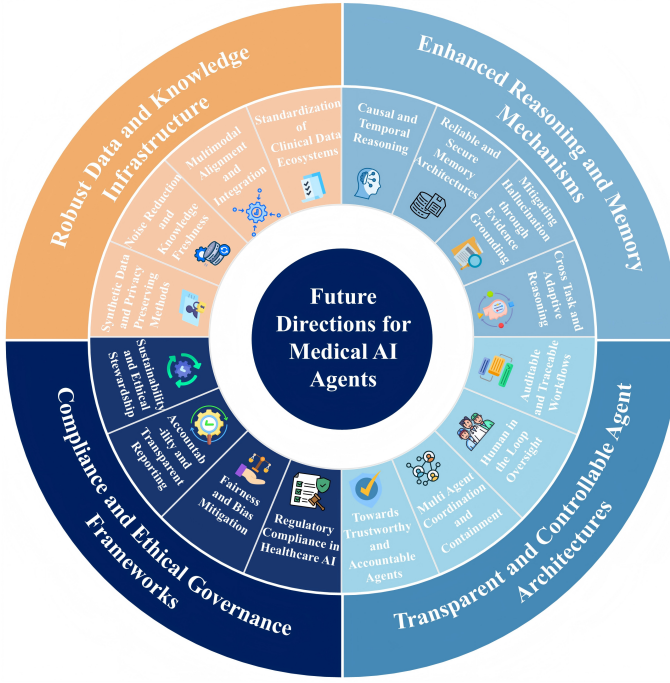


Fig. 5. Future directions for medical AI agents. The diagram organizes future research priorities into four domains: robust data and knowledge infrastructure, enhanced reasoning and memory mechanisms, transparent and controllable agent architectures, and compliance and ethical governance frameworks.

The absence of interpretability creates cascading effects. When errors occur, it is often impossible to determine whether they originated from flawed data, faulty reasoning, or hallucinated outputs. Such ambiguity delays correction and allows errors to propagate into downstream clinical decisions [132], [133]. For example, an agent suggesting an inappropriate drug dosage without offering traceable justification may cause the physician either to ignore potentially useful advice or, worse, to follow it blindly, putting patient safety at risk. These challenges emphasize the urgent need for medical AI agents to move beyond black box prediction and toward transparent, auditable reasoning frameworks [134]. Methods that can reveal intermediate reasoning steps, highlight supporting evidence, and quantify uncertainty are essential. Only by embedding explainability into the architecture of clinical agents can their recommendations become trustworthy, accountable, and aligned with the requirements of real-world medical practice [135].

3) *Clinical Deployment and Adoption*: One of the greatest barriers to clinical deployment of medical AI agents is the strict regulatory and ethical framework governing healthcare [136]. Unlike experimental domains, healthcare mandates compliance with regulations like HIPAA in the United States and GDPR in the European Union, which impose strict conditions on data access, processing, and sharing [137]. While crucial for patient privacy, these protections slow down the development and evaluation of AI systems [138]. Models cannot simply be trained on open datasets or updated in real time with patient records. Instead, each use case must pass

complex approval processes that delay adoption and restrict scalability.

The accountability challenge is magnified in high-stakes tasks like diagnosis, prescription, or surgical planning. In the absence of clear legal frameworks defining liability, hospitals and physicians are justifiably hesitant to rely on AI-driven recommendations [139]. If an error occurs, it remains ambiguous whether responsibility should fall on the developer, the healthcare provider, or the institution that deployed the system. This legal ambiguity creates hesitation among stakeholders and prevents agents from being trusted as integral components of patient care.

Ethical concerns, particularly regarding bias and fairness, further complicate deployment. Agents trained on non-representative data may perpetuate or even exacerbate healthcare disparities [140]. For example, a model trained predominantly on one demographic may yield less accurate recommendations for minority populations. The opacity of many systems can prevent clinicians from detecting these biased outcomes. Overcoming these obstacles requires the development of regulatory frameworks that balance innovation with safety, incorporating standardized approval processes, robust bias-auditing mechanisms, and transparent accountability pathways [141]. Aligning technological advancement with strong legal and ethical governance is paramount for transitioning medical AI agents from experimental trials to reliable, real-world clinical integration.

V. FUTURE DIRECTIONS

The next generation of medical AI agents will be shaped by advances across four interconnected domains: data infrastructure, reasoning mechanisms, agent architectures, and ethical governance. Fig. 5 provides a visual summary of these priorities, which we elaborate below.

A. Robust Data and Knowledge Infrastructure

1) *Standardization of Clinical Data Ecosystems*: A critical prerequisite for building robust infrastructures is the establishment of standardized and interoperable data ecosystems. Current healthcare data remain scattered across hospitals, insurance providers, and research institutes, each adopting its own formats for electronic health records, imaging archives, or genomic repositories. Even widely recognized standards such as HL7 and FHIR face uneven implementation, resulting in fragmented datasets that are difficult to merge [142], [143]. This lack of consistency directly prevents the construction of large-scale, patient-level corpora suitable for medical AI agents. A future research direction lies in developing unified data schemas and automated normalization pipelines that can harmonize heterogeneous inputs into a coherent representation. Such infrastructures would form the backbone of any scalable medical agent system [144].

2) *Multimodal Alignment and Integration*: Medical knowledge spans diverse modalities, including unstructured narratives, diagnostic images, laboratory records, and molecular profiles [145]. These sources are rarely aligned in practice,

with variations in annotation quality, resolution, and granularity. Existing multimodal fusion methods provide partial solutions but fail to address small sample problems such as rare disease cohorts. The challenge is not only technical but also systemic: many modalities are collected in isolated silos and cannot be easily linked to individual patient journeys. A robust infrastructure must therefore include multimodal alignment strategies that combine cross-modal embeddings, standardized ontologies, and graph-based integration frameworks. These approaches would allow agents to construct holistic views of patient states, enabling more accurate and generalizable reasoning [146].

3) *Noise Reduction and Knowledge Freshness*: Even when data can be accessed and aligned, quality remains a persistent issue. Clinical datasets are often noisy due to annotation errors, incomplete documentation, and inconsistent diagnostic guidelines [147]. These errors propagate into downstream reasoning, weakening the reliability of AI recommendations. Beyond noise, medical knowledge itself evolves rapidly. Drug safety updates, revised treatment guidelines, and newly classified diseases continuously reshape the clinical landscape [148]. Static corpora risk embedding outdated practices, leading agents to reproduce obsolete knowledge. Addressing these challenges requires dynamic pipelines for knowledge updating, incorporating retrieval augmentation, continual learning, and explicit version control. Such mechanisms would ensure that medical AI systems are not only accurate but also temporally aligned with current clinical standards.

4) *Synthetic Data and Privacy Preserving Methods*: Finally, the scarcity of high-quality annotated datasets and the restrictions imposed by privacy regulations create significant bottlenecks. Rare conditions and pediatric diseases provide too few training samples, while strict privacy laws prevent the large scale sharing of sensitive health records [149]. A promising direction is the development of synthetic data generation techniques, such as digital twins and simulation-based augmentation, which can expand training sets without compromising privacy. In parallel, federated learning and privacy preserving computation provide mechanisms for distributed training across institutions, enabling collaboration without direct data sharing [150]. Together, these innovations could alleviate the dual challenges of limited availability and regulatory constraints.

B. Enhanced Reasoning and Memory Mechanisms

1) *Causal and Temporal Reasoning*: One of the most pressing limitations of current medical AI agents lies in their reliance on statistical correlation rather than causal reasoning [151]. In clinical practice, patient outcomes are determined by complex cause and effect relationships, such as how comorbidities interact with treatments or how drug dosages influence long term prognosis. Without the ability to model causality, agents risk making brittle recommendations that fail under distributional shifts. Temporal reasoning is equally critical. Many clinical decisions depend on sequences of events, including progression of symptoms, longitudinal laboratory values, or treatment responses over time. Agents that cannot

capture these dependencies may produce recommendations that ignore crucial temporal context. Future research must therefore focus on integrating causal inference frameworks and temporal modeling into medical agents, enabling them to reason not only about what is correlated but also about what is mechanistically plausible.

2) *Reliable and Secure Memory Architectures*: Another fundamental challenge is the absence of robust long term memory mechanisms. Current systems operate with short contexts, leading to repeated errors and inconsistent reasoning across sessions. In healthcare, however, agents must recall prior interactions, maintain patient specific histories, and adapt to evolving treatment plans. At the same time, memory systems create new risks: sensitive information may be stored indefinitely or retrieved in unintended ways, exposing patients to privacy violations [152]. To address this dual challenge, future directions should focus on secure, auditable memory architectures. These architectures should support both retention and intentional forgetting, coupled with rigorous access control and data expiration policies. Such mechanisms would ensure continuity in reasoning while preserving patient confidentiality.

3) *Mitigating Hallucination through Evidence Grounding*: Medical AI agents are prone to hallucination, producing recommendations unsupported by evidence. In critical environments such as oncology or intensive care, these errors can propagate into severe consequences [153]. Hallucination becomes particularly dangerous when combined with memory persistence, as false information may be stored and reused in later interactions. Retrieval augmented generation and graph based reasoning offer promising solutions. By grounding outputs in curated biomedical knowledge bases, recent clinical guidelines, or validated databases, agents can provide recommendations that are both contextually accurate and verifiable. Combining retrieval with structured reasoning frameworks also enhances transparency, allowing clinicians to trace the evidential basis of agent decisions.

4) *Cross Task and Adaptive Reasoning*: Finally, clinical practice requires agents to move seamlessly across diverse reasoning tasks, from summarizing case notes to generating treatment options and verifying drug interactions. Current models lack the flexibility to coordinate such heterogeneous tasks, often leading to fragmented reasoning. Moreover, they struggle to generalize when faced with novel or rare conditions [154]. Future approaches should emphasize adaptive reasoning mechanisms, where agents dynamically adjust strategies based on task requirements, contextual uncertainty, and prior outcomes. Techniques such as meta reasoning, reflexive evaluation, and modular task orchestration hold promise in enabling agents to function as reliable collaborators in complex and evolving clinical environments.

C. Transparent and Controllable Agent Architectures

1) *Auditable and Traceable Workflows*: A fundamental obstacle in current medical AI agents is the absence of transparent workflows. Most agents operate as opaque systems where reasoning paths, tool calls, and memory updates are

hidden from clinical users. This opacity prevents physicians from auditing decisions and makes accountability difficult to enforce. In high risk environments, such as intensive care or oncology, the inability to reconstruct decision chains is a critical weakness [155]. Future work should focus on auditable agent pipelines, where each reasoning step, external query, and retrieved evidence is recorded in structured logs. These records would enable retrospective verification, support regulatory compliance, and provide clinicians with the ability to trace how recommendations were generated.

2) *Human in the Loop Oversight*: Another key limitation is the lack of mechanisms for human intervention. Current agents often act autonomously once deployed, leaving little room for clinicians to supervise, interrupt, or override outputs. This lack of controllability reduces physician trust and limits adoption in real world workflows. Transparent architectures must therefore include interactive control points that allow physicians to validate or correct agent reasoning in real time [156]. For instance, physicians should be able to approve or reject intermediate outputs before they propagate downstream. By embedding human in the loop mechanisms into the design, medical agents can be aligned more closely with clinical decision making practices and enhance overall safety.

3) *Multi-agent Coordination and Containment*: As medical AI shifts toward Multi-agent ecosystems, transparency and controllability become even more complex. Agents may exchange information asynchronously, increasing the risk of error propagation or uncontrolled behaviors. Without explicit coordination protocols, a single erroneous output can spread across the system and distort subsequent decisions. Future architectures should employ orchestration frameworks that define clear agent roles, permissions, and communication standards. Moreover, containment strategies are needed so that failures in one agent are isolated and do not cascade into systemic errors. Designing architectures with built in monitoring and arbitration will ensure that Multi-agent systems remain stable, interpretable, and clinically reliable.

4) *Towards Trustworthy and Accountable Agents*: The combination of auditability, human oversight, and structured coordination provides a path toward trustworthy medical AI agents. Transparent and controllable architectures will allow clinicians not only to observe how recommendations are generated but also to intervene when necessary. Such designs will align medical AI with regulatory expectations for accountability and safety, while also fostering clinician trust [157]. Ultimately, the success of medical agents in real deployments will depend not only on predictive performance but also on their ability to operate as accountable and controllable partners in clinical care.

D. Compliance and Ethical Governance Frameworks

1) *Regulatory Compliance in Healthcare AI*: Medical AI agents must operate under stringent legal frameworks such as HIPAA in the United States, GDPR in Europe, and the emerging EU AI Act. Current implementations rarely provide guarantees that recommendations, data storage, and model updates conform to these requirements [158]. Lack of compliance

mechanisms not only creates legal risks but also undermines the credibility of medical AI systems in real world deployments. Future designs must therefore integrate compliance by default, including secure data handling, traceable access logs, and mechanisms for continuous auditing aligned with national and international regulations.

2) *Fairness and Bias Mitigation*: Another pressing issue is the risk of amplifying inequities. Training data often reflect demographic imbalances, underrepresentation of rare conditions, or systemic biases in healthcare delivery [159]. Without explicit safeguards, AI agents may reproduce and even magnify these inequities in clinical practice. Future governance frameworks must incorporate fairness audits, bias detection pipelines, and benchmarks that systematically evaluate whether outputs are consistent across populations. Ensuring demographic diversity in both training and evaluation datasets will be essential to avoid reinforcing disparities in medical decision making.

3) *Accountability and Transparent Reporting*: Accountability is central to the safe adoption of medical AI agents. When an agent makes an erroneous recommendation, clinicians and patients must know who is responsible for the outcome and whether the error originated from data, model reasoning, or deployment protocols. Current black box systems obscure these causal chains [160]. Future governance should require transparent reporting mechanisms, including interpretable decision logs, uncertainty estimates, and responsibility assignments across the human–AI interaction loop. These measures will foster trust and allow errors to be identified, corrected, and prevented from recurring.

4) *Sustainability and Ethical Stewardship*: Beyond safety and fairness, the development of medical AI agents must also address sustainability and ethical stewardship. Large scale training consumes substantial computational resources, which translates into significant energy costs and environmental impact. Ethical governance frameworks should therefore prioritize efficiency, reproducibility, and low energy design, balancing innovation with sustainability [156]. In addition, governance structures must recognize the societal responsibilities of deploying medical AI, ensuring that benefits are equitably distributed and that vulnerable populations are not left behind in the adoption of advanced healthcare technologies.

VI. CONCLUSION

This survey has provided a comprehensive overview of the rapidly evolving field of AI agents in medicine, charting their progression from conceptual prototypes to sophisticated systems capable of tackling complex clinical tasks. Our central contribution is a novel, task-oriented taxonomy that organizes the landscape into four core domains: Clinical NLP, Medical Computer Vision, Graph Learning, and Omics and Biology. Through this framework, we have dissected the fundamental components, including planning, memory, and tool use, that enable agentic behavior. We have also identified and analyzed the critical challenges impeding widespread adoption, spanning data quality, clinical reasoning, and regulatory deployment. By synthesizing the current state of the art and

emerging trends, this survey offers a structured roadmap to guide future research and foster deeper collaboration between AI scientists and healthcare practitioners. The ultimate goal is to steer the development of the next generation of medical AI agents: systems that are not only more capable but also more reliable, transparent, and seamlessly integrated into clinical care to improve patient outcomes.

REFERENCES

- [1] L. Floridi and M. Chiriatti, “Gpt-3: Its nature, scope, limits, and consequences,” *Minds and machines*, vol. 30, no. 4, pp. 681–694, 2020.
- [2] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat *et al.*, “Gpt-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [3] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts, P. Barham, H. W. Chung, C. Sutton, S. Gehrmann *et al.*, “Palm: Scaling language modeling with pathways,” *Journal of Machine Learning Research*, vol. 24, no. 240, pp. 1–113, 2023.
- [4] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar *et al.*, “Llama: Open and efficient foundation language models,” *arXiv preprint arXiv:2302.13971*, 2023.
- [5] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang *et al.*, “Qwen technical report,” *arXiv preprint arXiv:2309.16609*, 2023.
- [6] C. Qian, X. Cong, C. Yang, W. Chen, Y. Su, J. Xu, Z. Liu, and M. Sun, “Communicative agents for software development,” *arXiv preprint arXiv:2307.07924*, vol. 6, no. 3, p. 1, 2023.
- [7] T. Abuelsaad, D. Akkil, P. Dey, A. Jagmohan, A. Vempaty, and R. Kokku, “Agent-e: From autonomous web navigation to foundational design principles in agentic systems,” *arXiv preprint arXiv:2407.13032*, 2024.
- [8] Z. Rasheed, M. Waseem, A. Ahmad, K.-K. Kemell, W. Xiaofeng, A. N. Duc, and P. Abrahamsson, “Can large language models serve as data analysts? a multi-agent assisted approach for qualitative data analysis,” *arXiv preprint arXiv:2402.01386*, 2024.
- [9] R. Elkamouchi, A. Daaif, and K. Elguemmat, “Multi-agents system in healthcare: A systematic literature review,” in *International Conference on Smart Applications and Data Analysis*. Springer, 2024, pp. 200–214.
- [10] Y. Inoue, T. Song, X. Wang, A. Luna, and T. Fu, “Drugagent: Multi-agent large language model-based reasoning for drug-target interaction prediction,” *ArXiv*, pp. ArXiv–2408, 2025.
- [11] X. Su, Y. Wang, S. Gao, X. Liu, V. Giunchiglia, D.-A. Clevert, and M. Zitnik, “Kgarevion: An ai agent for knowledge-intensive biomedical qa,” 2025.
- [12] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, and J. Gao, “Large language models: A survey,” *arXiv preprint arXiv:2402.06196*, 2024.
- [13] X. Huang, W. Liu, X. Chen, X. Wang, H. Wang, D. Lian, Y. Wang, R. Tang, and E. Chen, “Understanding the planning of llm agents: A survey,” *arXiv preprint arXiv:2402.02716*, 2024.
- [14] Z. Zhang, Q. Dai, X. Bo, C. Ma, R. Li, X. Chen, J. Zhu, Z. Dong, and J.-R. Wen, “A survey on the memory mechanism of large language model based agents,” *ACM Transactions on Information Systems*, 2024.
- [15] S. Yuan, K. Song, J. Chen, X. Tan, Y. Shen, R. Kan, D. Li, and D. Yang, “Easytool: Enhancing llm-based agents with concise tool instruction,” *arXiv preprint arXiv:2401.06201*, 2024.
- [16] X. Tang, A. Zou, Z. Zhang, Z. Li, Y. Zhao, X. Zhang, A. Cohan, and M. Gerstein, “MedAgents: Large language models as collaborators for zero-shot medical reasoning,” in *Findings of the Association for Computational Linguistics: ACL 2024*, L.-W. Ku, A. Martins, and V. Srikumar, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 599–621.
- [17] Y. Feng, J. Wang, L. Zhou, and Y. Li, “Doctoragent-rl: A multi-agent collaborative reinforcement learning system for multi-turn clinical dialogue,” *arXiv preprint arXiv:2505.19630*, 2025.
- [18] Z. Chen, M. Varma, J. Xu, M. Paschali, D. Van Veen, A. Johnston, A. Youssef, L. Blankemeier, C. Bluethgen, S. Altmayer *et al.*, “A vision-language foundation model to enhance efficiency of chest x-ray interpretation,” *arXiv e-prints*, pp. arXiv–2401, 2024.
- [19] D. Bani-Harouni, N. Navab, and M. Keicher, “Magda: Multi-agent guideline-driven diagnostic assistance,” in *International workshop on foundation models for general medical AI*. Springer, 2024, pp. 163–172.
- [20] G. Richard, B. P. de Almeida, H. Dalla-Torre, C. Blum, L. Hexemer, P. Pandey, S. Laurent, M. Lopez, A. Laterre, M. Lang *et al.*, “Chatnt: A multimodal conversational agent for dna, rna and protein tasks,” *bioRxiv*, pp. 2024–04, 2024.
- [21] Y. Huang, I. Cohen, V. Q.-T. Truong, P. B. Bayat, S. A. Bhatti, L. Paruzzo, M. M. Painter, S. Zheng, D. A. Oldridge, J. Wagenaar *et al.*, “Reference-free cell-type annotation with llm agents,” in *ICLR 2025 Workshop on Machine Learning for Genomics Explorations*.
- [22] J. Li, Y. Lai, W. Li, J. Ren, M. Zhang, X. Kang, S. Wang, P. Li, Y.-Q. Zhang, W. Ma *et al.*, “Agent hospital: A simulacrum of hospital with evolvable medical agents,” *arXiv preprint arXiv:2405.02957*, 2024.
- [23] Z. Yao and H. Yu, “A survey on llm-based multi-agent ai hospital,” 2025.
- [24] S. Ren, P. Jian, Z. Ren, C. Leng, C. Xie, and J. Zhang, “Towards scientific intelligence: A survey of llm-based scientific agents,” *arXiv preprint arXiv:2503.24047*, 2025.
- [25] N. Mehandru, B. Y. Miao, E. R. Almaraz, M. Sushil, A. J. Butte, and A. Alaa, “Evaluating large language models as agents in the clinic,” *NPJ digital medicine*, vol. 7, no. 1, p. 84, 2024.
- [26] Z. Fan, J. Tang, W. Chen, S. Wang, Z. Wei, J. Xi, F. Huang, and J. Zhou, “Ai hospital: Benchmarking large language models in a multi-agent medical interaction simulator,” *arXiv preprint arXiv:2402.09742*, 2024.
- [27] Y. Jiang, K. C. Black, G. Geng, D. Park, J. Zou, A. Y. Ng, and J. H. Chen, “Medagentbench: A realistic virtual ehr environment to benchmark medical llm agents,” *arXiv preprint arXiv:2501.14654*, 2025.
- [28] C. Xiaolan, X. Jiayang, L. Shanfu, L. Yexin, H. Mingguang, and S. Danli, “Evaluating large language models and agents in healthcare: key challenges in clinical applications,” *Intelligent Medicine*, 2025.
- [29] Y. Yang, H. Chai, Y. Song, S. Qi, M. Wen, N. Li, J. Liao, H. Hu, J. Lin, G. Chang *et al.*, “A survey of ai agent protocols,” *arXiv preprint arXiv:2504.16736*, 2025.
- [30] R. Sapkota, K. I. Roumeliotis, and M. Karkee, “Ai agents vs. agentic ai: A conceptual taxonomy, applications and challenges,” *arXiv preprint arXiv:2505.10468*, 2025.
- [31] T. Han, L. C. Adams, J.-M. Papaioannou, P. Grundmann, T. Oberhauser, A. Löser, D. Truhn, and K. K. Bressen, “Medalpaca—an open-source collection of medical conversational ai models and training data,” *arXiv preprint arXiv:2304.08247*, 2023.
- [32] T. Tu, S. Azizi, D. Driess, M. Schaeckermann, M. Amin, P.-C. Chang, A. Carroll, C. Lau, R. Tanno, I. Ktena *et al.*, “Towards generalist biomedical ai,” *Nejm Ai*, vol. 1, no. 3, p. A10a2300138, 2024.
- [33] N. Karunanayake, “Next-generation agentic ai for transforming healthcare,” *Informatics and Health*, vol. 2, no. 2, pp. 73–83, 2025.
- [34] K. Zhang, Y. Kang, F. Zhao, and X. Liu, “Llm-based medical assistant personalization with short-and long-term memory coordination,” *arXiv preprint arXiv:2309.11696*, 2023.
- [35] W. Wang, Z. Ma, Z. Wang, C. Wu, J. Ji, W. Chen, X. Li, and Y. Yuan, “A survey of llm-based agents in medicine: How far are we from baymax?” *arXiv preprint arXiv:2502.11211*, 2025.
- [36] X. Li, “A review of prominent paradigms for llm-based agents: Tool use, planning (including rag), and feedback learning,” in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 9760–9779.
- [37] S. A. Gebreab, K. Salah, R. Jayaraman, M. H. ur Rehman, and S. Ellaham, “Llm-based framework for administrative task automation in healthcare,” in *2024 12th International Symposium on Digital Forensics and Security (ISDFS)*. IEEE, 2024, pp. 1–7.
- [38] Y. Kim, C. Park, H. Jeong, Y. S. Chan, X. Xu, D. McDuff, H. Lee, M. Ghassemi, C. Breazeal, and H. W. Park, “Mdagents: An adaptive collaboration of llms for medical decision-making,” in *Advances in Neural Information Processing Systems*, A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, Eds., vol. 37. Curran Associates, Inc., 2024, pp. 79 410–79 452.
- [39] Z. Wang, Y. Zhu, H. Zhao, X. Zheng, D. Sui, T. Wang, W. Tang, Y. Wang, E. Harrison, C. Pan *et al.*, “Colacare: Enhancing electronic health record modeling through large language model-driven multi-agent collaboration,” in *Proceedings of the ACM on Web Conference 2025*, 2025, pp. 2250–2261.
- [40] M. Lu, B. Ho, D. Ren, and X. Wang, “Triageagent: Towards better multi-agents collaborations for large language model-based clinical

- triage,” in *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024, pp. 5747–5764.
- [41] H. Yu, J. Zhou, L. Li, S. Chen, J. Gallifant, A. Shi, X. Li, W. Hua, M. Jin, G. Chen *et al.*, “Aipatient: Simulating patients with ehers and llm powered agentic workflow,” *arXiv preprint arXiv:2409.18924*, 2024.
 - [42] Z. Liu, Q. Tu, W. Ye, Y. Xiao, Z. Zhang, H. Cui, Y. Zhu, Q. Ju, S. Li, and J. Xie, “Exploring the inquiry-diagnosis relationship with advanced patient simulators,” 2025.
 - [43] Z. Du, L. LujieZheng, R. Hu, Y. Xu, X. Li, Y. Sun, W. Chen, J. Wu, H. Cai, and H. Ying, “LLMs can simulate standardized patients via agent coevolution,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Vienna, Austria: Association for Computational Linguistics, Jul. 2025, pp. 17 278–17 306.
 - [44] X. Chen, Y. Jin, X. Mao, L. Wang, S. Zhang, and T. Chen, “Rareagents: Advancing rare disease care through llm-empowered multi-disciplinary team,” 2025.
 - [45] Y. Feng, J. Wang, L. Zhou, Z. Lei, and Y. Li, “Doctoragent-rl: A multi-agent collaborative reinforcement learning system for multi-turn clinical dialogue,” 2025.
 - [46] K. Chen, X. Li, T. Yang, H. Wang, W. Dong, and Y. Gao, “Mdeamgpt: A self-evolving llm-based multi-agent framework for multi-disciplinary team medical consultation,” 2025.
 - [47] H. Wang, S. Zhao, Z. Qiang, N. Xi, B. Qin, and T. Liu, “Beyond direct diagnosis: Llm-based multi-specialist agent consultation for automatic diagnosis,” *arXiv preprint arXiv:2401.16107*, 2024.
 - [48] W. Yan, H. Liu, T. Wu, Q. Chen, W. Wang, H. Chai, J. Wang, W. Zhao, Y. Zhang, R. Zhang, L. Zhu, and X. Zhao, “Clinicallab: Aligning agents for multi-departmental clinical diagnostics in the real world,” 2024. [Online]. Available: <https://arxiv.org/abs/2406.13890>
 - [49] M. Xiao, M. Ye, B. Liu, X. Zong, H. Li, J. Huang, Q. Xie, and M. Peng, “A retrieval-augmented multi-agent framework for psychiatry diagnosis,” *arXiv preprint arXiv:2506.03750*, 2025.
 - [50] Y. Ke, R. Yang, S. A. Lie, T. X. Y. Lim, Y. Ning, I. Li, H. R. Abdullah, D. S. W. Ting, and N. Liu, “Mitigating cognitive biases in clinical decision-making through multi-agent conversations using large language models: Simulation study,” *Journal of Medical Internet Research*, vol. 26, 2024.
 - [51] X. Zhou, Y. Ren, Q. Zhao, D. Huang, X. Wang, T. Zhao, Z. Zhu, W. He, S. Li, Y. Xu *et al.*, “An llm-driven multi-agent debate system for mendelian diseases,” *arXiv preprint arXiv:2504.07881*, 2025.
 - [52] M. Abbasian, I. Azimi, A. M. Rahmani, and R. Jain, “Conversational health agents: A personalized llm-powered agent framework,” *arXiv preprint arXiv:2310.02374*, 2023.
 - [53] X. Dong, W. Zhu, H. Wang, X. Chen, P. Qiu, R. Yin, Y. Su, and Y. Wang, “Talk before you retrieve: Agent-led discussions for better rag in medical qa,” *arXiv preprint arXiv:2504.21252*, 2025.
 - [54] D. Rose, C.-C. Hung, M. Lepri, I. Alqassem, K. Gashteovski, and C. Lawrence, “Meddxagent: A unified modular agent framework for explainable automatic differential diagnosis,” 2025.
 - [55] Z. Fan, J. Tang, W. Chen, S. Wang, Z. Wei, J. Xi, F. Huang, and J. Zhou, “Ai hospital: Benchmarking large language models in a multi-agent medical interaction simulator,” 2024.
 - [56] J. Wang, Z. Yao, Z. Yang, H. Zhou, R. Li, X. Wang, Y. Xu, and H. Yu, “NoteChat: A dataset of synthetic patient-physician conversations conditioned on clinical notes,” in *Findings of the Association for Computational Linguistics: ACL 2024*, L.-W. Ku, A. Martins, and V. Srikumar, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 15 183–15 201.
 - [57] S. Chen, M. Wu, K. Q. Zhu, K. Lan, Z. Zhang, and L. Cui, “Llm-empowered chatbots for psychiatrist and patient simulation: Application and evaluation,” 2023. [Online]. Available: <https://arxiv.org/abs/2305.13614>
 - [58] A. Dutta and Y.-C. Hsiao, “Adaptive reasoning and acting in medical language agents,” *arXiv preprint arXiv:2410.10020*, 2024.
 - [59] J. Liu, W. Wang, Z. Ma, G. Huang, Y. SU, K.-J. Chang, W. Chen, H. Li, L. Shen, and M. Lyu, “Medchain: Bridging the gap between llm agents and clinical practice through interactive sequential benchmarking,” *arXiv preprint arXiv:2412.01605*, 2024.
 - [60] V. Nair, E. Schumacher, G. Tso, and A. Kannan, “Dera: enhancing large language model completions with dialog-enabled resolving agents,” *arXiv preprint arXiv:2303.17071*, 2023.
 - [61] Z. Bao, Q. Liu, Y. Guo, Z. Ye, J. Shen, S. Xie, J. Peng, X. Huang, and Z. Wei, “Piors: Personalized intelligent outpatient reception based on large language model with multi-agents medical scenario simulation,” *arXiv preprint arXiv:2411.13902*, 2024.
 - [62] H. Wei, J. Qiu, H. Yu, and W. Yuan, “Medco: Medical education copilots based on a multi-agent framework,” in *Computer Vision – ECCV 2024 Workshops*, A. Del Bue, C. Canton, J. Pont-Tuset, and T. Tommasi, Eds. Cham: Springer Nature Switzerland, 2025, pp. 119–135.
 - [63] D. Bani-Harouni, C. Pellegrini, E. Özsoy, M. Keicher, and N. Navab, “Language agents for hypothesis-driven clinical decision making with reinforcement learning,” *arXiv preprint arXiv:2506.13474*, 2025.
 - [64] R. Xu, W. Shi, J. Wang, J. Zhou, and C. Yang, “Medassist: Llm-empowered medical assistant for assisting the scrutinization and comprehension of electronic health records,” in *Companion Proceedings of the ACM on Web Conference 2025*, 2025, pp. 2931–2934.
 - [65] S. Xu, Y. Zhou, Z. Liu, Z. Wu, T. Zhong, H. Zhao, Y. Li, H. Jiang, Y. Pan, J. Chen *et al.*, “Towards next-generation medical agent: How o1 is reshaping decision-making in medical scenarios,” *arXiv preprint arXiv:2411.14461*, 2024.
 - [66] D. Yang, J. Wei, M. Li, J. Liu, L. Liu, M. Hu, J. He, Y. Ju, W. Zhou, Y. Liu, and L. Zhang, “Medaide: Information fusion and anatomy of medical intents via llm-based agent collaboration,” 2025.
 - [67] D. Rose, C.-C. Hung, M. Lepri, I. Alqassem, K. Gashteovski, and C. Lawrence, “Meddxagent: A unified modular agent framework for explainable automatic differential diagnosis,” *arXiv preprint arXiv:2502.19175*, 2025.
 - [68] R. Xu, Y. Zhuang, Y. Zhong, Y. Yu, X. Tang, H. Wu, M. D. Wang, P. Ruan, D. Yang, T. Wang *et al.*, “Medagentgym: Training llm agents for code-based medical reasoning at scale,” *arXiv preprint arXiv:2506.04405*, 2025.
 - [69] F. Xu, P. Cheng, F. Gao, Y. Jin, S. Yan, Q. Huang, Y. Wang, X. Ren, and J. Gu, “An agent-based adaptive medical dialogue service for personalized healthcare,” *International Journal of Web Services Research (IJWSR)*, vol. 22, no. 1, pp. 1–28, 2025.
 - [70] X. Su, Y. Wang, S. Gao, X. Liu, V. Giunchiglia, D.-A. Clevert, and M. Zitnik, “Kgarevion: an ai agent for knowledge-intensive biomedical qa,” *arXiv preprint arXiv:2410.04660*, 2024.
 - [71] D. Bani-Harouni, N. Navab, and M. Keicher, “Magda: Multi-agent guideline-driven diagnostic assistance,” 2024.
 - [72] Z. Wang, J. Wu, L. Cai, C. H. Low, X. Yang, Q. Li, and Y. Jin, “Medagent-pro: Towards evidence-based multi-modal medical diagnosis via reasoning agentic workflow,” *arXiv preprint arXiv:2503.18968*, 2025.
 - [73] A. Hoopes, V. I. Butoi, J. V. Gutttag, and A. V. Dalca, “Voxelprompt: A vision-language agent for grounded medical image analysis,” 2024. [Online]. Available: <https://arxiv.org/abs/2410.08397>
 - [74] A. Fallahpour, J. Ma, A. Munim, H. Lyu, and B. Wang, “Medrax: Medical reasoning agent for chest x-ray,” 2025.
 - [75] F. Ghezloo, M. S. Seyfioglu, R. Soraki, W. O. Ikezogwo, B. Li, T. Vivekanandan, J. G. Elmore, R. Krishna, and L. Shapiro, “Pathfinder: A multi-modal multi-agent system for medical diagnostic decision-making applied to histopathology,” 2025.
 - [76] J. Feng, Q. Zheng, C. Wu, Z. Zhao, Y. Zhang, Y. Wang, and W. Xie, “M’ 3builder: A multi-agent system for automated machine learning in medical imaging,” *arXiv preprint arXiv:2502.20301*, 2025.
 - [77] J. Huang, R. He, D. Z. Khan, E. Mazomenos, D. Stoyanov, H. J. Marcus, M. J. Clarkson, and M. Islam, “Surgicalvlm-agent: Towards an interactive ai co-pilot for pituitary surgery,” 2025.
 - [78] Y. Mao, W. Xu, Y. Qin, and Y. Gao, “Ct-agent: A multimodal-llm agent for 3d ct radiology question answering,” 2025.
 - [79] P. Yang and B. Dong, “Mocoll: Agent-based specific and general model collaboration for image captioning,” *arXiv preprint arXiv:2501.01834*, 2025.
 - [80] Z. Yi, T. Xiao, and M. V. Albert, “A multimodal multi-agent framework for radiology report generation,” 2025.
 - [81] P. Xia, J. Wang, Y. Peng, K. Zeng, X. Wu, X. Tang, H. Zhu, Y. Li, S. Liu, Y. Lu, and H. Yao, “Mmedagent-rl: Optimizing multi-agent collaboration for multimodal medical reasoning,” 2025.
 - [82] F. M. Calisto, J. Fernandes, M. Morais, C. Santiago, J. M. Abrantes, N. Nunes, and J. C. Nascimento, “Assertiveness-based agent communication for a personalized medicine on medical imaging diagnosis,” in *Proceedings of the 2023 CHI conference on human factors in computing systems*, 2023, pp. 1–20.
 - [83] L. Zhao, J. Bai, Z. Bian, Q. Chen, Y. Li, G. Li, M. He, H. Yao, and Z. Zhang, “Autonomous multi-modal llm agents for treatment planning in focused ultrasound ablation surgery,” *arXiv preprint arXiv:2505.21418*, 2025.
 - [84] B. Li, T. Yan, Y. Pan, J. Luo, R. Ji, J. Ding, Z. Xu, S. Liu, H. Dong, Z. Lin *et al.*, “Mmedagent: Learning to use medical tools with multi-modal agent,” *arXiv preprint arXiv:2407.02483*, 2024.

- [85] P. Wang, Z. Zhang, G. Gao, Y. Zhang, and Z. Zheng, "Agentpolyp: Accurate polyp segmentation via image enhancement agent," *IEEE Signal Processing Letters*, no. 99, pp. 1–5, 2025.
- [86] C. H. Low, Z. Wang, T. Zhang, Z. Zeng, Z. Zhuo, E. B. Mazomenos, and Y. Jin, "Surgraw: Multi-agent workflow with chain-of-thought reasoning for surgical intelligence," 2025.
- [87] J. Jin and C. W. Jeong, "Surgical-llava: Toward surgical scenario understanding via large language and vision models," *arXiv preprint arXiv:2410.09750*, 2024.
- [88] Z. Yi, J. Liu, T. Xiao, and M. V. Albert, "A multi-agent system for complex reasoning in radiology visual question answering," *arXiv preprint arXiv:2508.02841*, 2025.
- [89] R. Zhang, Y. Sun, Z. Zhang, J. Li, X. Liu, A. H. Fan, H. Guo, and P. Yan, "Marl-mambacontour: Unleashing multi-agent deep reinforcement learning for active contour optimization in medical image segmentation," *arXiv preprint arXiv:2506.18679*, 2025.
- [90] U. Cho and N. Kim, "A-ide: Agent-integrated denoising experts," *arXiv preprint arXiv:2503.16780*, 2025.
- [91] B. Yang, S. Jiang, L. Xu, K. Liu, H. Li, G. Xing, H. Chen, X. Jiang, and Z. Yan, "Drhouse: An llm-empowered diagnostic reasoning system through harnessing outcomes from sensor data and expert knowledge," 2024.
- [92] C. Chen, L. L. Weishaupt, D. F. Williamson, R. J. Chen, T. Ding, B. Chen, A. Vaidya, L. P. Le, G. Jaume, M. Y. Lu *et al.*, "Evidence-based diagnostic reasoning with multi-agent copilot for human pathology," *arXiv preprint arXiv:2506.20964*, 2025.
- [93] P. R. Liu, S. Bansal, J. Dinh, A. Pawar, R. Satishkumar, S. Desai, N. Gupta, X. Wang, and S. Hu, "Medchat: A multi-agent framework for multimodal diagnosis with large language models," *arXiv preprint arXiv:2506.07400*, 2025.
- [94] W. Hou, G. Yang, Y. Du, Y. Lau, L. Liu, J. He, L. Long, and S. Wang, "Adagent: Llm agent for alzheimer's disease analysis with collaborative coordinator," *arXiv preprint arXiv:2506.11150*, 2025.
- [95] Y. Zhou, L. Song, and J. Shen, "Mam: Modular multi-agent framework for multi-modal medical diagnosis via role-specialized collaboration," *arXiv preprint arXiv:2506.19835*, 2025.
- [96] Y. Zhou, P. Zhang, M. Song, A. Zheng, Y. Lu, Z. Liu, Y. Chen, and Z. Xi, "Zodiac: A cardiologist-level llm framework for multi-agent diagnostics," *arXiv preprint arXiv:2410.02026*, 2024.
- [97] Y. Sun, Y. Si, C. Zhu, K. Zhang, Z. Shui, B. Ding, T. Lin, and L. Yang, "Cpathagent: An agent-based foundation model for interpretable high-resolution pathology image analysis mimicking pathologists' diagnostic logic," *arXiv preprint arXiv:2505.20510*, 2025.
- [98] G. A. Sajua, M. Akhib, and Y. Chang, "Agentmri: A vision language model-powered ai system for self-regulating mri reconstruction with multiple degradations," *Journal of Imaging Informatics in Medicine*, pp. 1–19, 2025.
- [99] K. Zuo, Y. Jiang, F. Mo, and P. Lio, "Kg4diagnosis: A hierarchical multi-agent llm framework with knowledge graph enhancement for medical diagnosis," in *Proceedings of The First AAAI Bridge Program on AI for Medicine and Healthcare*, ser. Proceedings of Machine Learning Research, J. Wu, J. Zhu, M. Xu, and Y. Jin, Eds., vol. 281. PMLR, 25 Feb 2025, pp. 195–204.
- [100] Y. Lu and J. Wang, "Karma: Leveraging multi-agent llms for automated knowledge graph enrichment," 2025.
- [101] J. A. Dehkordi, M. S. Parsa, M. Naghipourfar, and M. Mofrad, "Koda: An agentic framework for kegg orthology-driven discovery of antimicrobial drug targets in gut microbiome," in *ICML 2025 Generative AI and Biology (GenBio) Workshop*, 2025.
- [102] S. Gao, A. Fang, Y. Huang, V. Giunchiglia, A. Noori, J. R. Schwarz, Y. Ektefaie, J. Kondic, and M. Zitnik, "Empowering biomedical discovery with ai agents," *Cell*, vol. 187, no. 22, pp. 6125–6151, 2024.
- [103] K. Song, A. Trotter, and J. Y. Chen, "Llm agent swarm for hypothesis-driven drug discovery," *arXiv preprint arXiv:2504.17967*, 2025.
- [104] N. Lee, E. De Brouwer, E. Hajiramezanali, T. Biancalani, C. Park, and G. Scalia, "Rag-enhanced collaborative llm agents for drug discovery," *arXiv preprint arXiv:2502.17506*, 2025.
- [105] N. Matsumoto, H. Choi, J. Moran, M. E. Hernandez, M. Venkatesan, X. Li, J.-H. Chang, P. Wang, and J. H. Moore, "Escargot: an ai agent leveraging large language models, dynamic graph of thoughts, and biomedical knowledge graphs for enhanced reasoning," *Bioinformatics*, vol. 41, no. 2, p. btaf031, 01 2025.
- [106] K. Zuo, Z. Zhong, P. Huang, S. Tang, Y. Chen, and Y. Jiang, "Heal-kggen: A hierarchical multi-agent llm framework with knowledge graph enhancement for genetic biomarker-based medical diagnosis," *bioRxiv*, 2025.
- [107] X. Lin, S. Ma, J. Shan, X. Zhang, S. X. Hu, T. Guo, S. Z. Li, and K. Yu, "Biokgbench: A knowledge graph checking benchmark of ai agent for biomedical science," *arXiv preprint arXiv:2407.00466*, 2024.
- [108] L. Yue, S. Xing, J. Chen, and T. Fu, "Clinicalagent: Clinical trial multi-agent system with large language model-based reasoning," 2024. [Online]. Available: <https://arxiv.org/abs/2404.14777>
- [109] H. Shi, J. Zhang, and K. Zhang, "Enhancing clinical trial patient matching through knowledge augmentation and reasoning with multi-agent," 2025. [Online]. Available: <https://arxiv.org/abs/2411.14637>
- [110] X. Zhao, M. Blum, R. Yang, B. Yang, L. M. Carpintero, M. Pina-Navarro, T. Wang, X. Li, H. Li, Y. Fu *et al.*, "Agentigraph: An interactive knowledge graph platform for llm-based chatbots utilizing private data," *arXiv preprint arXiv:2410.11531*, 2024.
- [111] N. Lee, E. D. Brouwer, E. Hajiramezanali, T. Biancalani, C. Park, and G. Scalia, "Rag-enhanced collaborative llm agents for drug discovery," 2025.
- [112] G. Singh, L. Wehling, A. W. Mulyadi, R. H. Sreenath, T. Klabunde, T. Andreani, and D. McCloskey, "Talk2biomodels and talk2knowledgegraph: Ai agent-based application for prediction of patient biomarkers and reasoning over biomedical knowledge graphs," in *ICLR 2025 Workshop on Machine Learning for Genomics Explorations*, 2025. [Online]. Available: <https://openreview.net/forum?id=a4vQhBNezO>
- [113] K. Huang, S. Zhang, H. Wang, Y. Qu, Y. Lu, Y. Roohani, R. Li, L. Qiu, G. Li, J. Zhang, D. Yin, S. Marwaha, J. N. Carter, X. Zhou, M. Wheeler, J. A. Bernstein, M. Wang, P. He, J. Zhou, M. Snyder, L. Cong, A. Regev, and J. Leskovec, "Biomni: A general-purpose biomedical ai agent," 2025.
- [114] Y. Roohani, A. Lee, Q. Huang, J. Vora, Z. Steinhart, K. Huang, A. Marson, P. Liang, and J. Leskovec, "Biodiscoveryagent: An ai agent for designing genetic perturbation experiments," 2025. [Online]. Available: <https://arxiv.org/abs/2405.17631>
- [115] Y. Shen, Z. Chen, M. Mamalakis, Y. Liu, T. Li, Y. Su, J. He, P. Liò, and Y. G. Wang, "Toursynbio: A multi-modal large model and agent framework to bridge text and protein sequences for protein engineering," 2024. [Online]. Available: <https://arxiv.org/abs/2408.15299>
- [116] Z. Wang, Q. Jin, C.-H. Wei, S. Tian, P.-T. Lai, Q. Zhu, C.-P. Day, C. Ross, and Z. Lu, "Geneagent: Self-verification language agent for gene set knowledge discovery using domain databases," 2024. [Online]. Available: <https://arxiv.org/abs/2405.16205>
- [117] K. Swanson, W. Wu, N. L. Bulaong, J. E. Pak, and J. Zou, "The virtual lab of ai agents designs new sars-cov-2 nanobodies," *Nature*, pp. 1–3, 2025.
- [118] B. Gao, Y. Huang, Y. Liu, W. Xie, W.-Y. Ma, Y.-Q. Zhang, and Y. Lan, "Pharmagents: Building a virtual pharma with large language model agents," *arXiv preprint arXiv:2503.22164*, 2025.
- [119] G. V. Solovev, A. B. Zhidkovskaya, A. Orlova, A. Vepreva, T. Ilya, R. Golovinskii, N. Gubina, D. Chistiakov, T. A. Aliev, I. Poddiaikov, G. Zubkova, E. V. Skorb, V. Vinogradov, N. Nikitin, A. Dmitrenko, A. Kalyuzhnaya, and A. Savchenko, "Towards LLM-driven multi-agent pipeline for drug discovery: Neurodegenerative diseases case study," in *2nd AI4Research Workshop: Towards a Knowledge-grounded Scientific Research Lifecycle*, 2024. [Online]. Available: <https://openreview.net/forum?id=3ncjySu5ro>
- [120] I. Stewart and M. J. Buehler, "Molecular analysis and design using generative artificial intelligence via multi-agent modeling," *Mol. Syst. Des. Eng.*, vol. 10, pp. 314–337, 2025.
- [121] Y. Pu, T. Lin, and H. Chen, "Piflow: Principle-aware scientific discovery with multi-agent collaboration," 2025. [Online]. Available: <https://arxiv.org/abs/2505.15047>
- [122] H. Liu, S. Chen, Y. Zhang, and H. Wang, "Genotex: An llm agent benchmark for automated gene expression data analysis," 2025. [Online]. Available: <https://arxiv.org/abs/2406.15341>
- [123] Y. Xiao, E. Sun, Y. Jin, Q. Wang, and W. Wang, "Proteingpt: Multimodal llm for protein property prediction and structure understanding," 2025. [Online]. Available: <https://arxiv.org/abs/2408.11363>
- [124] A. Ghafarollahi and M. J. Buehler, "Protagents: Protein discovery via large language model multi-agent collaborations combining physics and machine learning," 2024. [Online]. Available: <https://arxiv.org/abs/2402.04268>
- [125] M. Hao, Y. Lee, H. Wang, G. Scalia, and A. Regev, "Perturboagent: A self-planning agent for boosting sequential perturb-seq experiments," *bioRxiv*, pp. 2025–05, 2025.
- [126] D. Bender and K. Sartipi, "HI7 fhir: An agile and restful approach to healthcare information exchange," in *Proceedings of the 26th IEEE*

- International Symposium on Computer-Based Medical Systems*, 2013, pp. 326–331.
- [127] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, “Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases,” in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3462–3471.
 - [128] D. P. Germain, D. Gruson, M. Malcles, and N. Garcelon, “Applying artificial intelligence to rare diseases: a literature review highlighting lessons from fabry disease,” *Orphanet Journal of Rare Diseases*, vol. 20, no. 1, p. 186, 2025.
 - [129] S. S. Band, A. Yarahmadi, C.-C. Hsu, M. Biyari, M. Sookhak, R. Ameri, I. Dehzaangi, A. T. Chronopoulos, and H.-W. Liang, “Application of explainable artificial intelligence in medical health: A systematic review of interpretability methods,” *Informatics in Medicine Unlocked*, vol. 40, p. 101286, 2023.
 - [130] S. Ramgopal, L. N. Sanchez-Pinto, C. M. Horvat, M. S. Carroll, Y. Luo, and T. A. Florin, “Artificial intelligence-based clinical decision support in pediatrics,” *Pediatric Research*, vol. 93, no. 2, pp. 334–341, 2023.
 - [131] A. Rao, K. Pal, S. Singh, D. Kunwar, S. N. Kumar, P. K. Sahu, T. Sharma, E. Shehzad, D. Kumar, R. K. Verma, P. Agrawal, and A. Goel, “Assessing the utility of chatgpt throughout the entire clinical workflow: Development and usability study,” *Journal of Medical Internet Research*, vol. 25, no. 1, p. e48659, 2023.
 - [132] Z. Sadeghi, R. Alizadehsani, M. A. CIFCI, S. Kausar, R. Rehman, P. Mahanta, P. K. Bora, A. Almasri, R. S. Alkhalwaldeh, S. Hussain, B. Alatas, A. Shoeibi, H. Moosaei, M. Hladik, S. Nahavandi, and P. M. Pardalos, “A review of explainable artificial intelligence in healthcare,” *Computers and Electrical Engineering*, vol. 118, p. 109370, 2024.
 - [133] J. Amann, A. Blasimme, E. Vayena, D. Frey, and V. I. Madai, “Explainability for artificial intelligence in healthcare: a multidisciplinary perspective,” *BMC Medical Informatics and Decision Making*, vol. 20, no. 1, p. 310, 2020.
 - [134] L. Huang, S. Ruan, Y. Xing, and M. Feng, “A review of uncertainty quantification in medical image analysis: probabilistic and non-probabilistic methods,” *Artificial Intelligence in Medicine*, vol. 136, p. 102152, 2024.
 - [135] M. Salvi, S. Seoni, A. Campagner, A. Gertych, U. Acharya, F. Molinari, and F. Cabitza, “Explainability and uncertainty: Two sides of the same coin for enhancing the interpretability of deep learning models in healthcare,” *International Journal of Medical Informatics*, vol. 197, p. 105846, 2025.
 - [136] S. Benjamens, P. Dhunoo, and B. Meskó, “The state of artificial intelligence-based fda-approved medical devices and algorithms: an online database,” *npj Digital Medicine*, vol. 3, p. 118, 2020.
 - [137] —, “The state of artificial intelligence-based fda-approved medical devices and algorithms: an online database,” *npj Digital Medicine*, vol. 3, p. 118, 2020.
 - [138] H. Li, M. Zhang, R. Gao, and et al., “Metabolomic adaptations and correlates of survival to immune-checkpoint blockade,” *Nature Medicine*, vol. 25, no. 12, p. 1458–1466, 2019.
 - [139] D. L. Longo and J. M. Drazen, “Data sharing,” *N Engl J Med*, vol. 378, no. 13, pp. 1281–1283, 2019.
 - [140] B. M. Knosp, J. Campion, T. R., N. Keating, H. Mori, J. Patsy, L. Swan, C. Wald et al., “Understanding enterprise data warehouses to support clinical and translational research,” *Journal of the American Medical Informatics Association*, vol. 29, no. 4, p. 671–676, 2022.
 - [141] M. Taddeo, “Cyber conflicts and political power in information societies,” *Minds and Machines*, vol. 27, no. 2, pp. 265–268, 2017.
 - [142] E. Williams and a. et, “A standardized clinical data harmonization pipeline for clinical data sets relying on the common fhir data standard,” *JAMIA Open*, vol. 6, no. 4, pp. 575–582, 2023.
 - [143] M. Ayaz, M. F. Pasha, M. Y. Alzahrani, R. Budiarto, and D. Stiawan, “The fast health interoperability resources (fhir) standard: Systematic literature review of implementations, applications, challenges and opportunities,” *JMIR Medical Informatics*, vol. 9, no. 7, p. e21929, 2021.
 - [144] I. Bossenko, R. Randmaa, G. Piho, and P. Ross, “Interoperability of health data using fhir mapping language: transforming hl7 cda to fhir with reusable visual components,” *Frontiers in Digital Health*, vol. 6, p. 1480600, 2024.
 - [145] J. N. Acosta, G. J. Falcone, P. Rajpurkar, and E. J. Topol, “Multimodal biomedical ai,” *Nature Medicine*, vol. 28, no. 9, pp. 1773–1784, 2022.
 - [146] T. Shaik, X. Tao, L. Li, H. Xie, and J. D. Velásquez, “A survey of multimodal information fusion for smart healthcare: Mapping the journey from data to wisdom,” 2023.
 - [147] A. Syloypavan, D. Sleeman, H. Wu, and M. Sim, “The impact of inconsistent human annotations on ai driven clinical decision making,” *npj Digital Medicine*, vol. 6, no. 1, p. 26, 2023.
 - [148] J. Feng, R. V. Phillips, I. Malenica, A. Bishara, A. E. Hubbard, L. A. Celi, and R. Pirracchio, “Clinical artificial intelligence quality improvement: towards continual monitoring and updating of ai algorithms in healthcare,” *npj Digital Medicine*, vol. 5, no. 1, p. 66, 2022.
 - [149] M. Giuffrè and D. L. Shung, “Harnessing the power of synthetic data in healthcare: innovation, application, and privacy,” *npj Digital Medicine*, vol. 6, no. 1, p. 186, 2023.
 - [150] S. Pati, S. Kumar, A. Varma, B. Edwards, C. Lu, L. Qu, J. J. Wang, A. Lakshminarayanan, S.-H. Wang, M. J. Sheller, K. Chang, P. Singh, D. L. Rubin, J. Kalpathy-Cramer, and S. Bakas, “Privacy preservation for federated learning in health care,” *Patterns (N.Y.)*, vol. 5, no. 7, p. 100974, 2024.
 - [151] J. M. Smit, J. H. Krijthe, W. M. R. Kant, J. A. Labrecque, M. Komorowski, D. A. M. P. J. Gommers, J. van Bommel, M. J. T. Reinders, and M. E. van Genderen, “Causal inference using observational intensive care unit data: a scoping review and recommendations for future practice,” *npj Digital Medicine*, vol. 6, no. 1, p. 221, 2023.
 - [152] R. Yang, Y. Ning, E. Keppo, M. Liu, C. Hong, D. S. Bitterman, J. C. L. Ong, D. S. W. Ting, and N. Liu, “Retrieval-augmented generation for generative artificial intelligence in health care,” *npj Health Systems*, vol. 2, no. 1, p. 2, 2025.
 - [153] Y. Kim, H. Jeong, S. Chen, S. S. Li, M. Lu, K. Alhamoud, J. Mun, C. Grau, M. Jung, R. Gameiro, L. Fan, E. Park, T. Lin, J. Yoon, W. Yoon, M. Sap, Y. Tsvetkov, P. Liang, X. Xu, X. Liu, D. McDuff, H. Lee, H. W. Park, S. Tulebaev, and C. Breazeal, “Medical hallucinations in foundation models and their impact on healthcare,” 2025.
 - [154] H. Lee, H.-K. Yoon, J. Kim, J. S. Park, C.-H. Koo, D. Won, and H.-C. Lee, “Development and validation of a reinforcement learning model for ventilation control during emergence from general anesthesia,” *npj Digital Medicine*, vol. 6, no. 1, p. 145, 2023.
 - [155] D. S. Char, M. D. Abramoff, and C. Feudtner, “Implementing machine learning in health care — addressing ethical challenges,” *New England Journal of Medicine*, vol. 383, no. 10, pp. 981–983, 2020.
 - [156] E. Topol, “High-performance medicine: the convergence of human and artificial intelligence,” *Nature Medicine*, vol. 25, pp. 44–56, 2019.
 - [157] K. Kittichokechai et al., “Safety and trustworthiness in multi-agent systems for healthcare,” *IEEE Intelligent Systems*, vol. 38, no. 3, pp. 38–48, 2023.
 - [158] S. Gerke, T. Minssen, and I. G. Cohen, “Ethical and legal challenges of artificial intelligence-driven healthcare,” *Nature Medicine*, vol. 26, pp. 1461–1467, 2020.
 - [159] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science*, vol. 366, no. 6464, pp. 447–453, 2019.
 - [160] I. Habli, R. Lawton, J. Porter et al., “Artificial intelligence in health care: accountability and safety assurance,” *BMJ Health & Care Informatics*, vol. 27, p. e100106, 2020.